

(12) **United States Patent**  
**Starratt et al.**

(10) **Patent No.:** **US 12,407,757 B2**  
(45) **Date of Patent:** **\*Sep. 2, 2025**

(54) **METHODS AND SYSTEMS FOR IMPLEMENTING AN INTELLIGENT DIGITAL ASSISTANT IN A MULTI-APPLICATION NETWORK**

(71) Applicant: **Black Knight IP Holding Company, LLC**, Jacksonville, FL (US)

(72) Inventors: **Kara S. Starratt**, Jacksonville, FL (US); **David W. Denson**, Jacksonville, FL (US); **James A. Iredale**, Ponte Vedra Beach, FL (US); **Sandra T. Madigan**, Jacksonville, FL (US); **Erik J. Skinner**, Sturgeon Bay, WI (US); **Lesley Grimes**, Jacksonville, FL (US)

(73) Assignee: **Black Knight IP Holding Company, LLC**, Jacksonville, FL (US)

(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.  
  
This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/829,042**

(22) Filed: **Sep. 9, 2024**

(65) **Prior Publication Data**  
US 2025/0106288 A1 Mar. 27, 2025

**Related U.S. Application Data**  
(63) Continuation of application No. 18/373,797, filed on Sep. 27, 2023, now Pat. No. 12,088,673.  
(51) **Int. Cl.**  
**H04L 67/125** (2022.01)  
**G06F 40/30** (2020.01)  
(52) **U.S. Cl.**  
CPC ..... **H04L 67/125** (2013.01); **G06F 40/30** (2020.01)

(58) **Field of Classification Search**  
CPC ..... H04L 67/125; G06F 40/30  
See application file for complete search history.

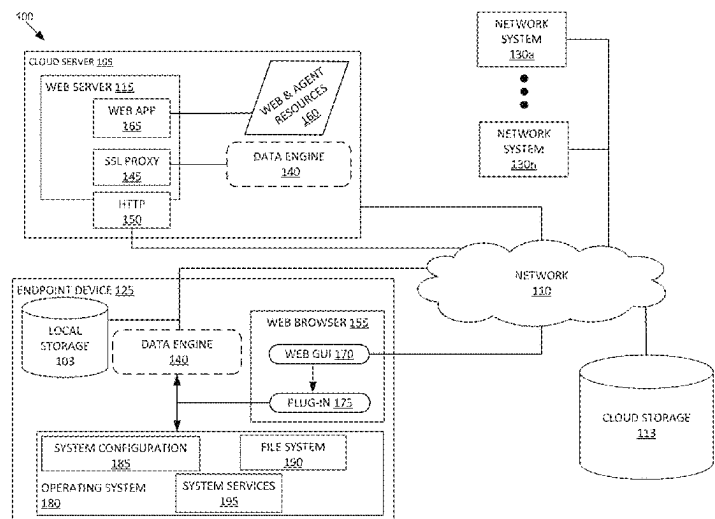
(56) **References Cited**  
**U.S. PATENT DOCUMENTS**  
  
6,434,628 B1 8/2002 Bowman-Amuah  
9,130,900 B2 9/2015 Tran  
(Continued)

**OTHER PUBLICATIONS**  
  
Non-Final Office Action dated Nov. 6, 2024 in connection with U.S. Appl. No. 18/829,037 34 pages.  
(Continued)

*Primary Examiner* — Soe Hlaing  
(74) *Attorney, Agent, or Firm* — DLA Piper LLP (US)

(57) **ABSTRACT**  
  
The disclosed method for implementing a digital assistant in a multi-application network includes: receiving an input command from a computing device; analyzing the input command; determining based on the analysis of the input command, a digital request data object associated with the input command; resolving the input command into a data string; determining, based on the digital request data object, intent data; and generating based on the intent data, a first set of operation recommendations comprising a first operation recommendation associated with a first application function. The methods also include determining an application programming interface (API) associated with the first application function; connecting, using the API, to the first application function; executing the first operation recommendation to generate a computing operation result; and rendering the computing operation on a single graphical interface comprising a consolidation of a plurality of graphical interfaces associated with the first set of computing operation recommendations.

**23 Claims, 7 Drawing Sheets**



(56)

**References Cited**

## U.S. PATENT DOCUMENTS

9,830,187 B1 11/2017 Blaine et al.  
 10,068,301 B2 9/2018 Kogut-O'Connell et al.  
 10,810,361 B1 10/2020 Venkatraman et al.  
 10,963,316 B2 3/2021 Gopalan et al.  
 11,379,268 B1 7/2022 Gupta  
 11,379,897 B1 7/2022 Thomas  
 12,050,592 B1 7/2024 Starratt et al.  
 12,061,596 B1 8/2024 Starratt et al.  
 12,088,673 B1 9/2024 Starratt et al.  
 12,135,978 B1\* 11/2024 Starratt ..... G06F 9/453  
 2002/0002453 A1 1/2002 Lazaridis et al.  
 2002/0078432 A1 6/2002 Charisius et al.  
 2002/0138449 A1 9/2002 Kendall et al.  
 2003/0233374 A1 12/2003 Spinola et al.  
 2004/0215552 A1 10/2004 Horn et al.  
 2005/0028073 A1 2/2005 Henry et al.  
 2005/0165734 A1 7/2005 Vicars et al.  
 2006/0069596 A1 3/2006 Hatoun et al.  
 2006/0069605 A1 3/2006 Hatoun et al.  
 2006/0074714 A1 4/2006 Aziz et al.  
 2008/0114791 A1 5/2008 Takatsu et al.  
 2009/0089869 A1 4/2009 Varghese  
 2009/0276340 A1 11/2009 Knapp  
 2010/0050153 A1 2/2010 Louie et al.  
 2010/0070464 A1 3/2010 Aymeloglu et al.  
 2011/0055180 A1 3/2011 Lumley et al.  
 2011/0282707 A1 11/2011 Rangaswamy et al.  
 2012/0236201 A1 9/2012 Larsen et al.  
 2013/0152021 A1 6/2013 Hatfield et al.  
 2013/0212151 A1 8/2013 Herbach et al.  
 2014/0047028 A1 2/2014 Buth  
 2014/0351115 A1 11/2014 Dahiwardkar  
 2015/0134600 A1 5/2015 Eisner et al.  
 2016/0104067 A1 4/2016 Xu et al.  
 2018/0121217 A1 5/2018 Jarabek et al.  
 2018/0196683 A1\* 7/2018 Radebaugh ..... G06F 3/167  
 2019/0268462 A1 8/2019 Mm et al.  
 2020/0151630 A1 5/2020 Shakhnovich  
 2020/0320407 A1 10/2020 Xiao et al.  
 2020/0341834 A1 10/2020 Safary et al.  
 2020/0372088 A1 11/2020 Liu et al.  
 2020/0394612 A1 12/2020 Khokhar et al.  
 2020/0410395 A1 12/2020 Ray et al.  
 2021/0089375 A1 3/2021 Ghafourifar et al.  
 2021/0192134 A1 6/2021 Yue et al.  
 2022/0237567 A1\* 7/2022 Tiwari ..... G06Q 10/1053  
 2022/0358286 A1 11/2022 Wilson-Thomas et al.  
 2023/0004460 A1 1/2023 Nagar et al.  
 2023/0066058 A1 3/2023 Yadav et al.  
 2023/0106058 A1 4/2023 Lavery et al.  
 2023/0119035 A1 4/2023 Mee

2023/0125194 A1 4/2023 Mertens et al.  
 2023/0134235 A1 5/2023 Setlur et al.  
 2023/0188480 A1\* 6/2023 Menon ..... G06N 5/041  
 709/206

2023/0368198 A1 11/2023 Gonzales, Jr.  
 2023/0395067 A1 12/2023 Perkins et al.

## OTHER PUBLICATIONS

Final Office Action dated May 22, 2024 in connection with U.S. Appl. No. 18/373,822, 25 pages.  
 Oracle, "What is a Digital Assistant?", Feb. 8, 2023 snapshot via Archive.org, URL Link: <<https://www.oracle.com/chatbots/what-is-a-digital-assistant/>> Accessed Nov. 2023 (Year: 2023), 3 pages.  
 Workgrid, "What is a Digital Assistant?", Jun. 1, 2023 snapshot via Archive.org, URL Link: <<https://workgrid.com/article/what-is-a-digital-assistant/>> Accessed Nov. 2023 (Year: 2023), 2 pages.  
 Qiao Huang, API Method Recommendation without Worrying about the Task-API Knowledge Gap. (Year: 2018).  
 TurboTax User Guide, URL Link: <[https://www.goladderup.org/wp-content/uploads/2020/03/IRS\\_Free\\_File\\_by\\_TurboTax\\_Guide\\_for\\_2019.pdf](https://www.goladderup.org/wp-content/uploads/2020/03/IRS_Free_File_by_TurboTax_Guide_for_2019.pdf)>, (Year: 2019), 21 pages.  
 Shareef, How to Set up Microsoft's Remote Desktop Connection, URL Link: <<https://www.makeuseof.com/how-to-set-up-microsofts-remote-desktop-connection/>>, (Year: 2021), 18 pages.  
 Honda service Bulletin, URL Link: <<https://static.nhtsa.gov/odi/tsbs/2023/MC-10230496-0001.pdf>>, (Year: 2023), 23 pages.  
 Non-Final Office Action dated Nov. 22, 2023 in connection with U.S. Appl. No. 18/373,830, 30 pages.  
 Non-Final Office Action dated Nov. 22, 2023 in connection with U.S. Appl. No. 18/373,771, 27 pages.  
 Non-Final Office Action dated Dec. 11, 2023 in connection with U.S. Appl. No. 18/373,813, 31 pages.  
 Non-Final Office Action dated Dec. 20, 2023 in connection with U.S. Appl. No. 18/373,797, 22 pages.  
 Non-Final Office Action dated Jan. 31, 2024 in connection with U.S. Appl. No. 18/373,822, 23 pages.  
 Notice of Allowance dated Mar. 4, 2024 in connection with U.S. Appl. No. 18/373,771, 27 pages.  
 Notice of Allowance dated Mar. 19, 2024 in connection with U.S. Appl. No. 18/373,830, 19 pages.  
 Notice of Allowance dated Mar. 28, 2024 in connection with U.S. Appl. No. 18/373,813, 22 pages.  
 Notice of Allowance dated Apr. 10, 2024 in connection with U.S. Appl. No. 18/373,797, 24 pages.  
 Non-Final Office Action dated Dec. 16, 2024 in connection with U.S. Appl. No. 18/885,191 32 pages.  
 Notice of Allowance dated Feb. 11, 2025 in connection with U.S. Appl. No. 18/885,191, 7 pages.

\* cited by examiner

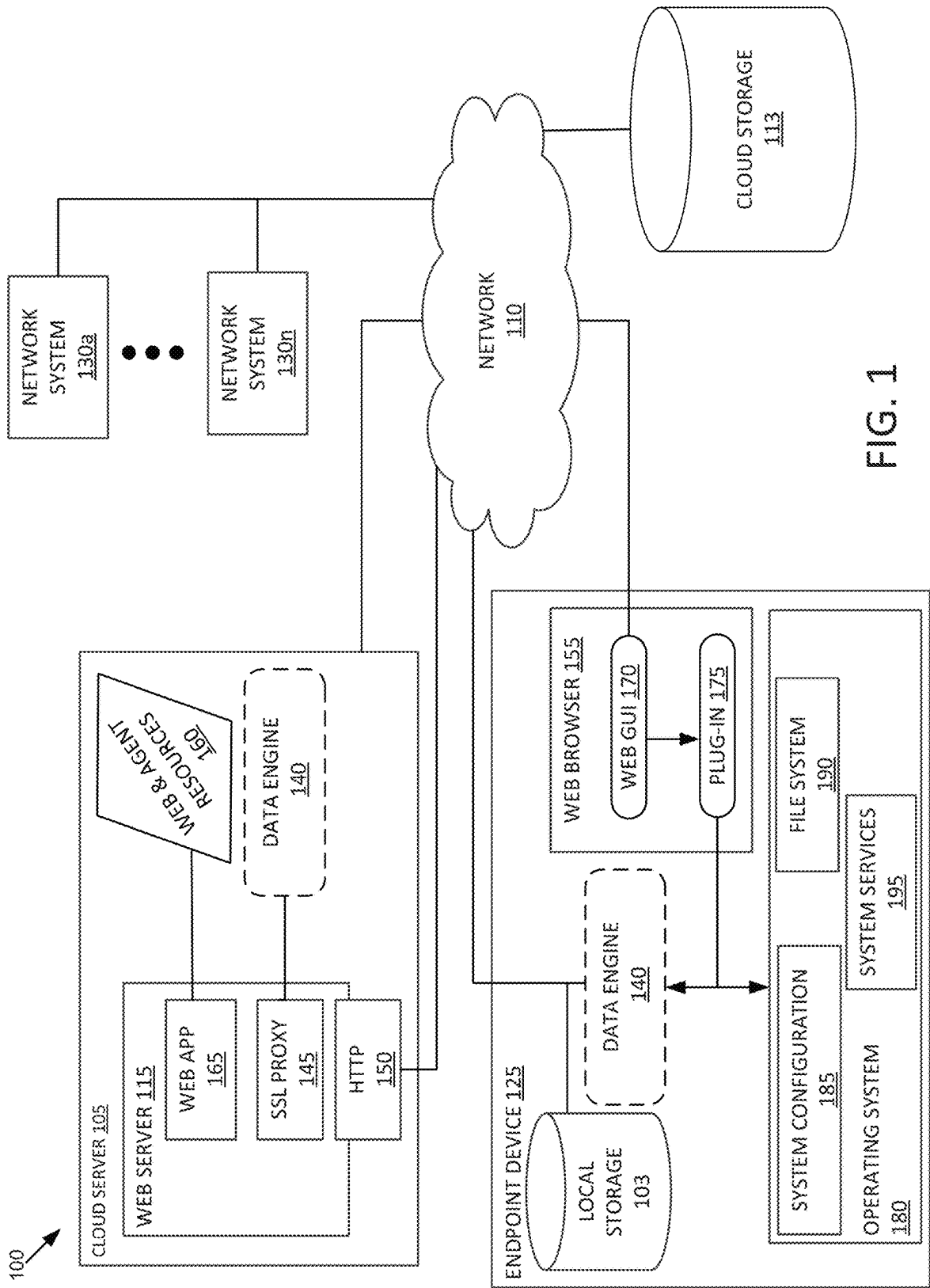


FIG. 1

200 →

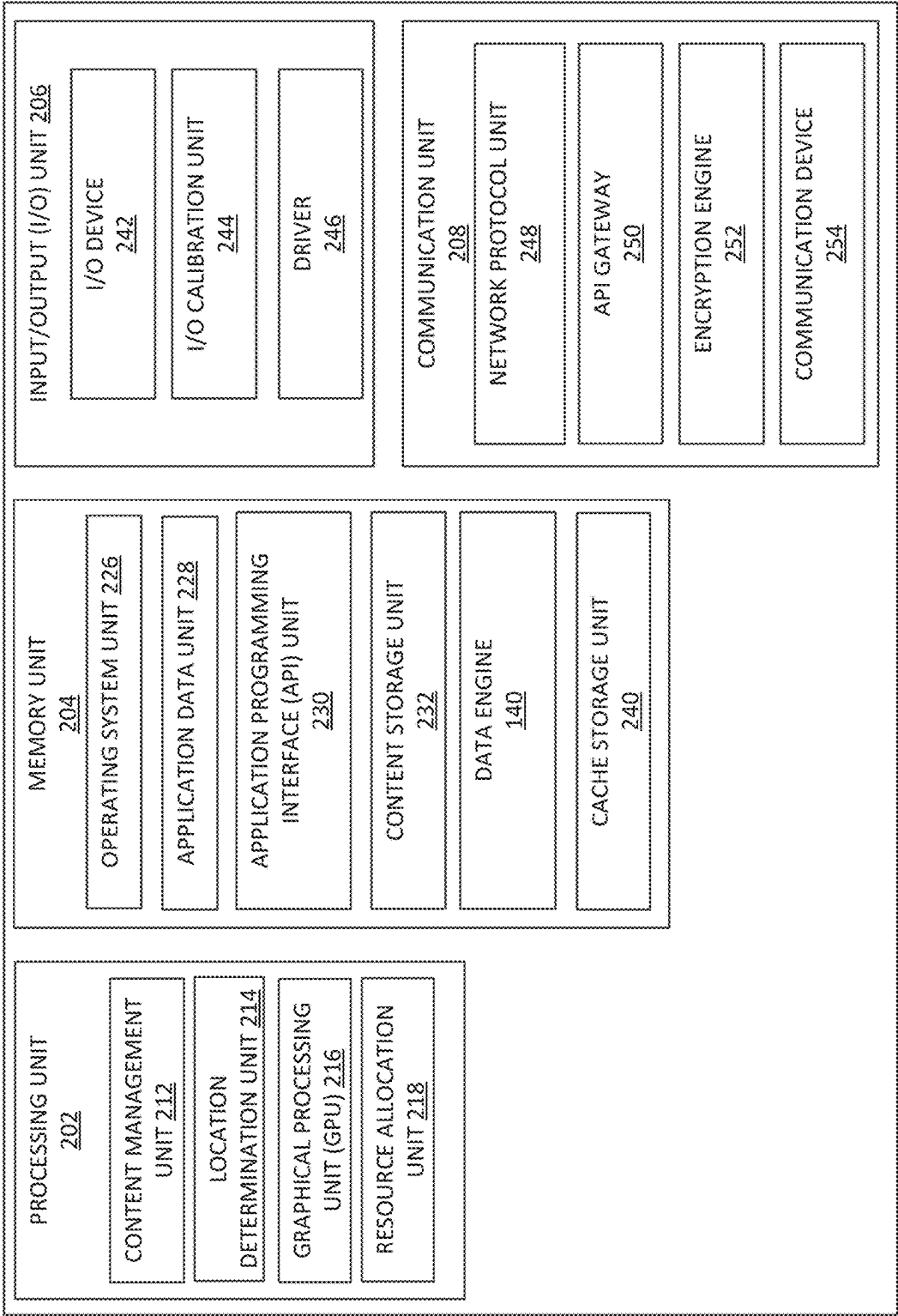


FIG. 2

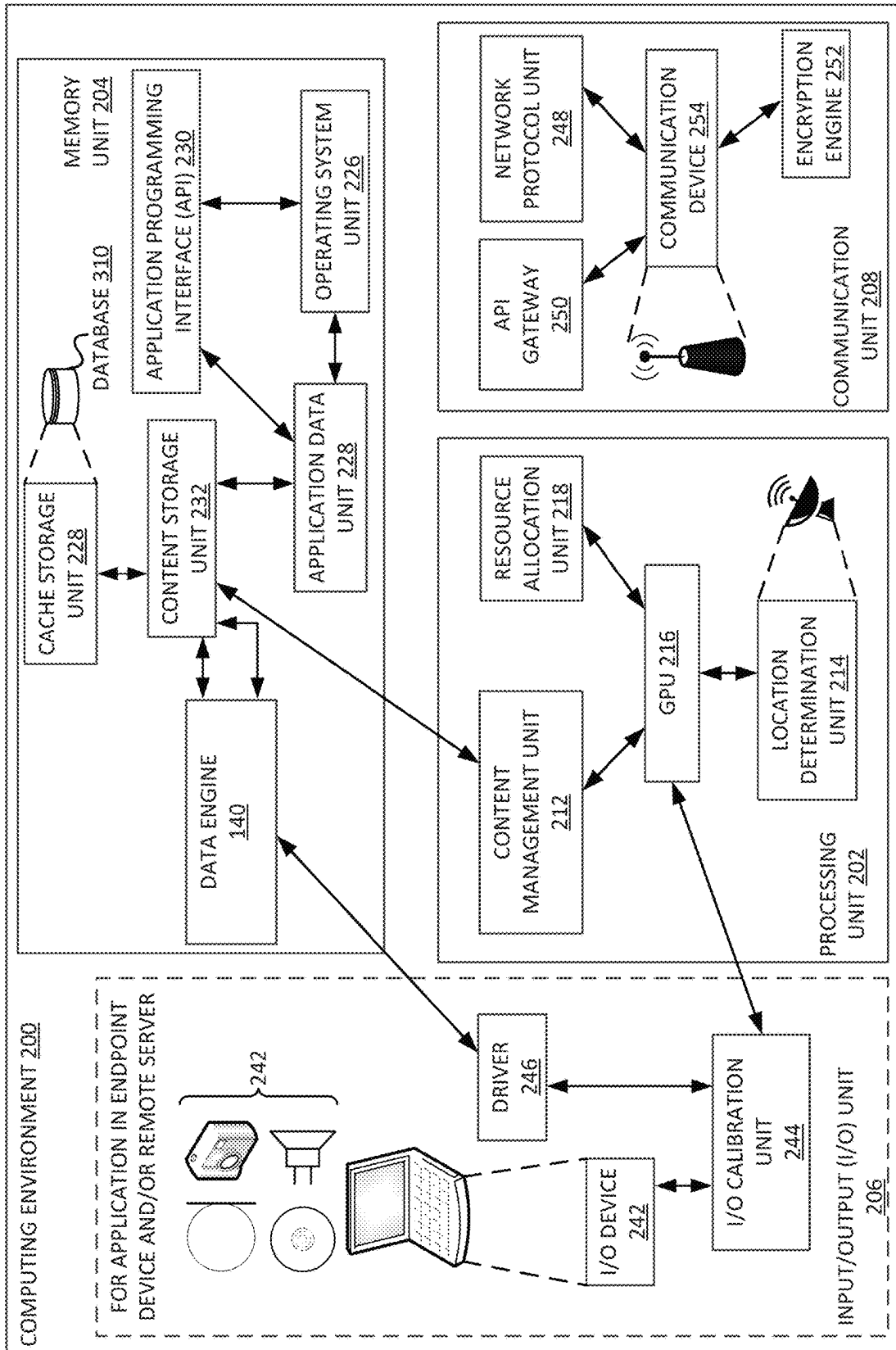


FIG. 3

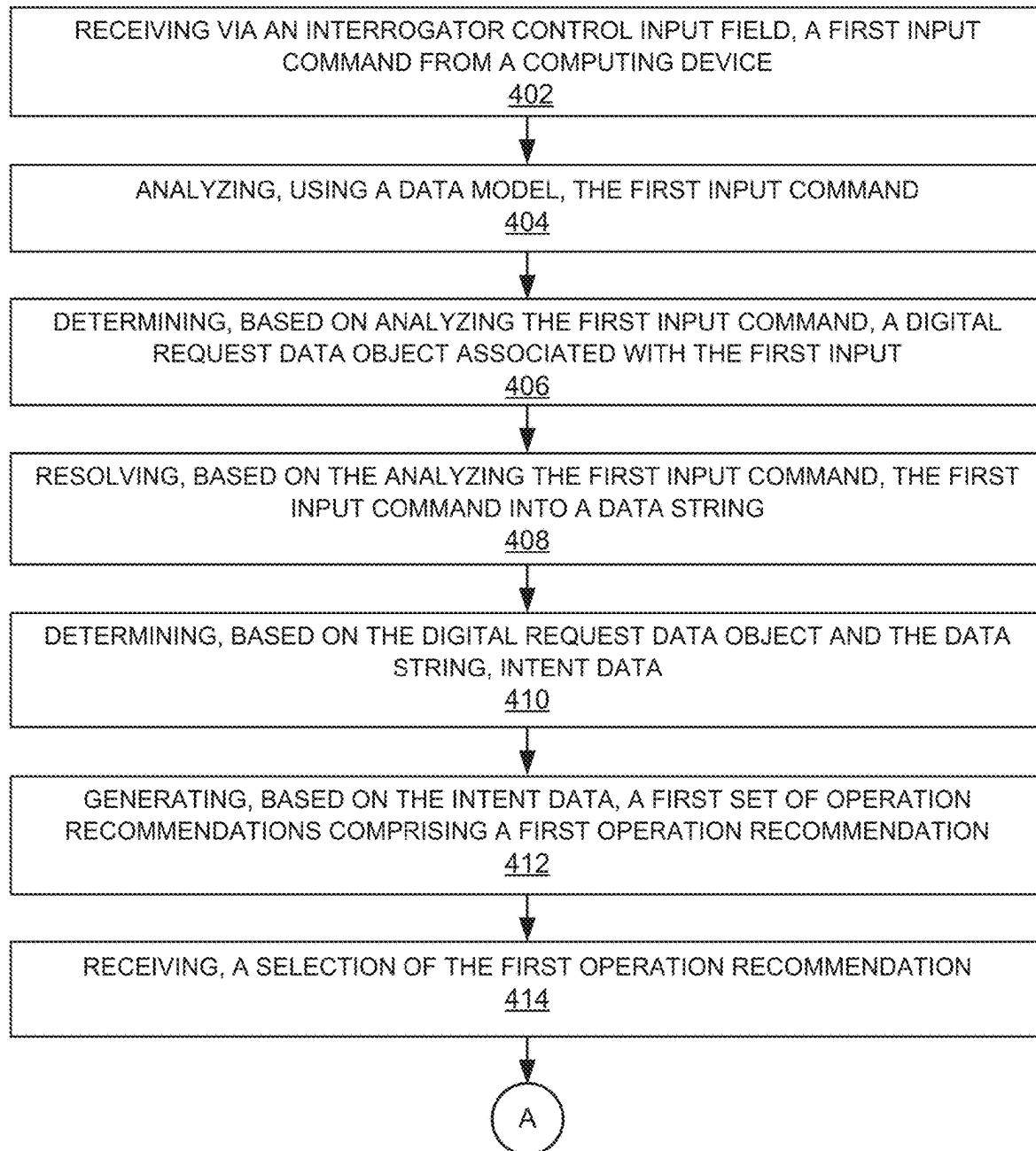


FIG. 4A

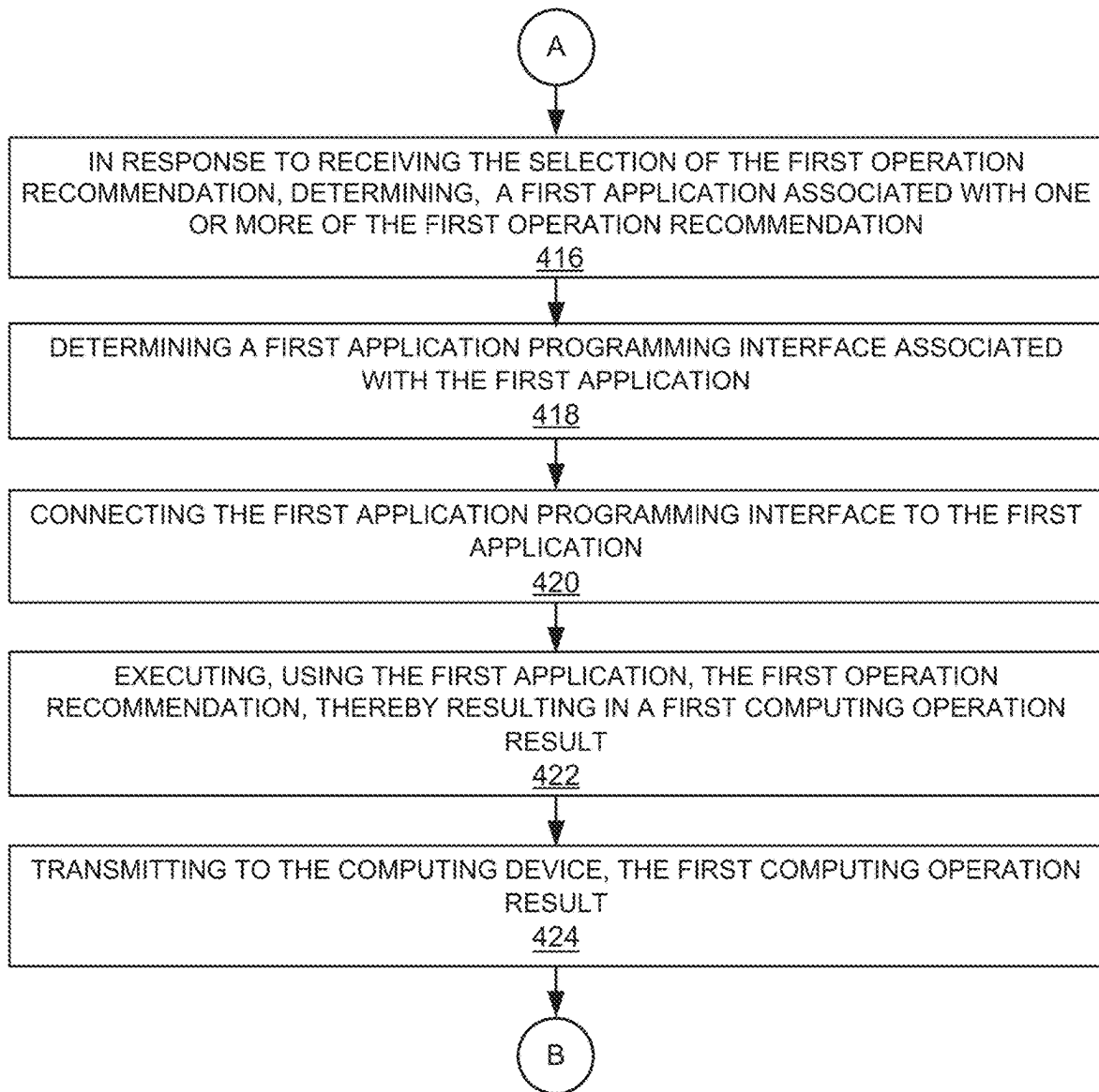


FIG. 4B

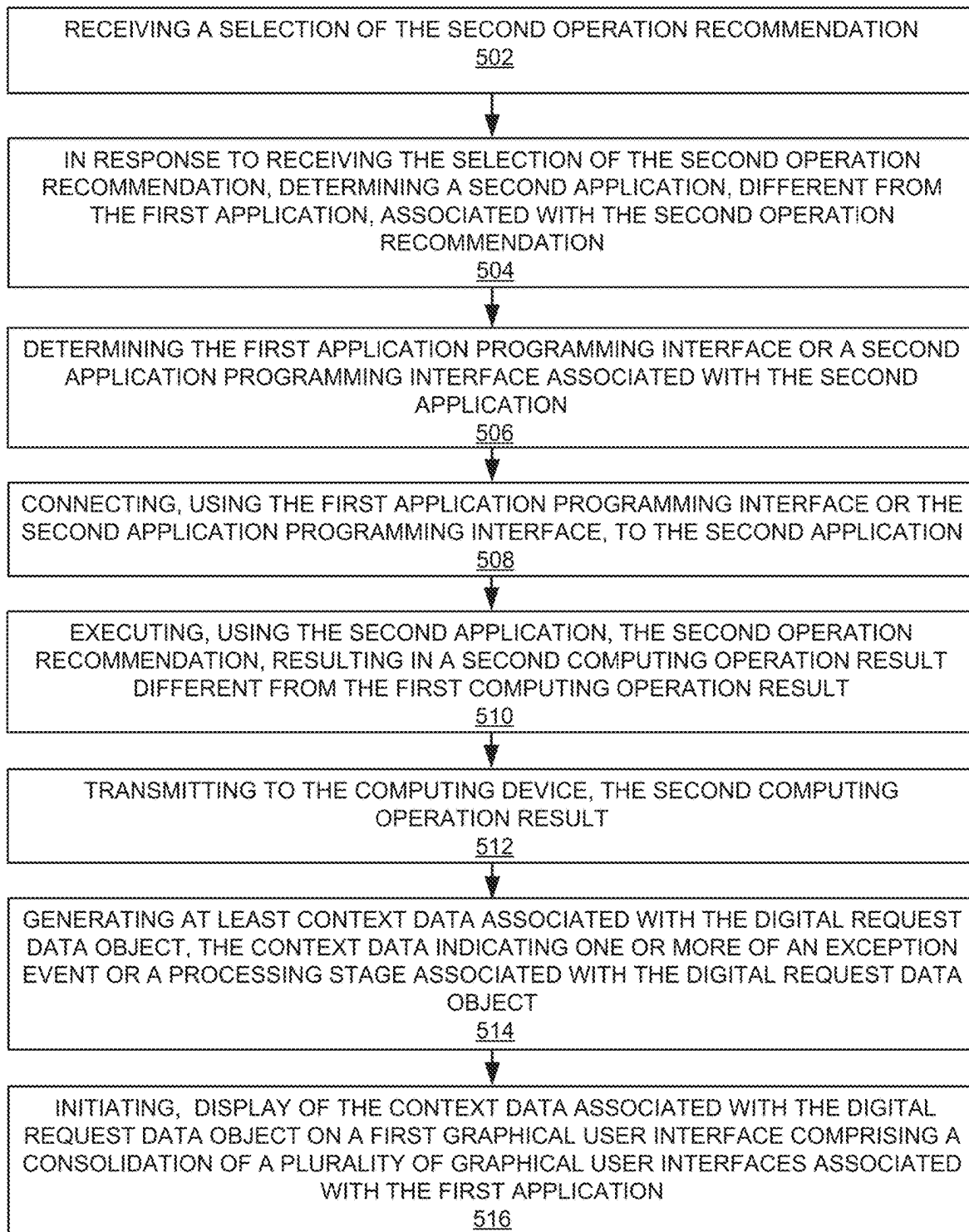


FIG. 5



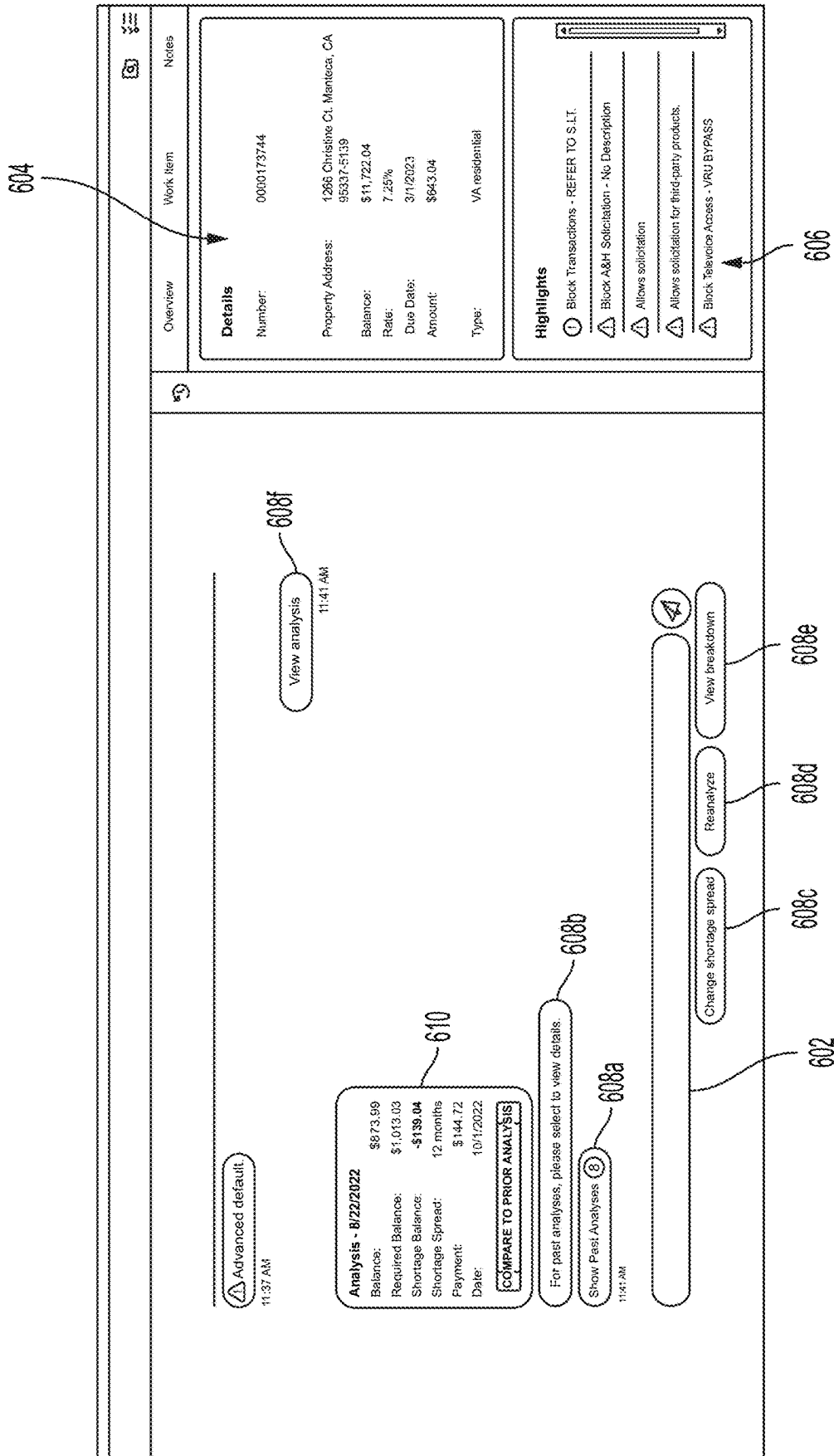


FIG. 6

1

# METHODS AND SYSTEMS FOR IMPLEMENTING AN INTELLIGENT DIGITAL ASSISTANT IN A MULTI-APPLICATION NETWORK

## CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 18/373,797 filed on Sep. 27, 2023, the disclosure of which is incorporated by reference in its entirety for all purposes.

## TECHNICAL FIELD

The present disclosure relates to methods, systems and computer program products that facilitate the implementation and usage of digital commands and a digital assistant in a multi-application network.

## BACKGROUND

Integrating and leveraging data from multiple applications and/or domains associated with a multi-application network is needed to facilitate process optimizations as well as efficient data cataloging, data tracking, exception event handling, data contextualization, and/or data storage. In particular, it is increasingly becoming needful to develop and leverage computational tools that efficiently register, activate, and/or deactivate commands within a multi-application network based on data associated with one or more digital request data objects associated with a multi-application network and/or other contextual data associated with the multi-application network, and/or user-specific, or group-specific data of users and/or groups that use the multi-application network.

## SUMMARY

The present disclosure is directed to methods, systems/apparatuses, and computer program products that implement the registering of a command in a multi-application network. The methods include determining a first computing operation for the multi-application network. The first computing operation, according to one embodiment, may be associated with a first digital request data object. For example, the first computing operation may comprise: a status check of the digital request data object; a review of at least one exception event associated with the digital request data object; quantitative and/or qualitative analysis of the first digital request data object; etc. The methods may further comprise receiving a first command for the first computing operation associated with the first digital request data object. The methods further include encapsulating, in a first registration object, one or more of: first semantic or syntactic data associated with one or more components of the first command for the computing operation associated with the first digital request data object; a digital path to a first application for executing the first computing operation; and context data associated with the first command for the first computing operation associated with the first digital request data object. The methods also include storing, in a database associated with the multi-application network, the first registration object. In addition, the methods include receiving, a first input associated with a second digital request data object. The first input may comprise textual data and/or auditory or vocal data received using an input device. In one embodi-

2

ment, the textual and/or auditory data may comprise natural language inputs with associated syntactic and semantic data (e.g., semantic and/or syntactic parameters) characterizing the first input. According to some embodiments, the semantic and/or syntactic data include a logical flow of one or more words comprised in the first input. Furthermore, the semantic and/or syntactic data may include data associated with an arrangement or organization of one or more words or text comprised in the first input. The methods may further include analyzing the first input to determine second semantic or syntactic data associated with the first input. The methods may further include determining the second semantic or syntactic data substantially matches the first semantic or syntactic data. In response to determining if the second semantic or syntactic data substantially matches the first semantic or syntactic data, the method proceeds to identifying the first command associated with the first computing operation. The methods may further include determining the digital path comprised in the first registration object. According to one embodiment, the methods include accessing, the first registration object based on the first command following which the digital path comprised in the first registration object is determined. Based on the digital path comprised in the first registration object, a first application for the first computing operation is accessed following the first computing operation is executed using the first application for the second digital request data object.

According to one embodiment, the methods include receiving a second input associated with a third digital request data object. The method may also include analyzing the second input to determine third semantic or syntactic data associated with the second input. In response to determining that the third semantic or syntactic data does not substantially match the first semantic or syntactic data, the methods proceed to determining or establishing that the second input is not executable for the third digital request data object based on the second input.

These and other implementations may each optionally include one or more of the following features. The first input associated with the digital request data object is a natural language input. In one embodiment, natural language may comprise a human language (e.g., English language, French language, Portuguese language, Chinese language, Japanese language, Korean language, Dutch language, etc.) that is spoken or written. Furthermore, analyzing the first input by the data engine may comprise: resolving the natural language input into one or more data strings; and applying the first semantic or syntactic data to the one or more data strings to generate the second semantic or syntactic data. In some embodiments, the digital request data object can comprise a file, or a document, or a record, or profile data associated with a user request (e.g., a digital request), or profile data associated with a digital service comprised in or associated with the multi-application network. The file, document, record, or profile data associated with the digital request data object, or profile data associated with a service comprised in, or associated with the multi-application network can outline or be associated with one or more of: account data associated with the digital request data object; or parametric data associated with resolving one or more exception events associated with the digital request data object. In addition, encapsulating the first registration object can comprise: mapping one or more metadata associated with the digital command to the context data to generate one or more registration identifiers; and including or incorporating the one or more registration identifiers in the first registration object.

Moreover, the first computing operation comprises a computing operation for determining an exception event associated with the digital request data object. For example, the exception event can indicate a completion status associated with a processing stage of the digital request data object.

In some implementations, the data engine may encapsulate a second registration object such that the second registration object comprises one or more of: fourth semantic or syntactic data associated with the one or more components of a second command; a digital path to a second application for executing a second computing operation associated with the second command; and context data associated with the second command. The second computing operation, for example, may comprise a computing operation for solving, addressing, or managing a detected exception event associated with the digital request data object.

Moreover, the first registration object may be activated, by the data engine for execution of the first computing operation based on one or more of: context data associated with the first input; device profile data associated with a first computing device through which the first input was received; and user profile data associated with a user providing the first input. Similarly, the first registration object may be deactivated by the data engine for execution of the first computing operation based on one or more of: context data associated with a second input; device profile data associated with a second computing device through which the second input was received; and user profile data associated with a user providing the second input.

According to one embodiment, the context data may control, confirm, or otherwise regulate: the formatting and/or presentation of analysis operation recommendations associated with the digital request data object based on the one or more user inputs; exception event detection operations associated with one or more of the digital request data object; recommendation of stage-wise progression of operations that resolve exception events associated with the digital request data object; and communicate with one or more APIs coupled to one or more native or non-native or third-party applications that execute workflows based on one or more of a user input (e.g., via a digital assistant), a selected analysis operation recommendation, or the context data (e.g., new or updated context data). In some embodiments, the digital assistant can match one or more keywords in a user input to one or more dynamically configured operations associated with a digital request data object to generate the one or more analysis operation recommendations for selection by the user. Furthermore, based on the context data, the digital assistant can suggest other tasks or data types to the user for selection as the case may require. If the digital assistant makes a suggestion that a user is not interested in, the user can provide other input commands (e.g., natural language input commands) clarifying the context data for the digital assistant to generate pertinent outputs for selection by the user. In such cases, the digital assistant (e.g., powered by an artificial intelligence engine) is able to track and assimilate a trajectory of user inputs leading to an eventual recommendation selection by the user and at a later date provide said recommendations for the same or other set of user inputs associated with the same or different digital request data objects.

In addition, the digital path of the first registration object can comprise a script that is executed to activate the first application (e.g., via an API) during accessing the first application. In some embodiments, the digital assistant is configured to receive: the first input; or the second input; or

a third input associated with resolving an exception event of the first digital request data object, or the second digital request data object.

Furthermore, a computing operation result may be generated, by the data engine, in response to executing the first computing operation. The computing operation result may comprise a first set of computing operation recommendations including a second computing operation associated with the digital request data object and which is executed by a second application. In addition, the computing operation result may be displayed on a single user interface that is configured to display a plurality of computing operation results generated from analysis operations executed by the first application or the second application. The single user interface may display the plurality of computing operation results instead of a plurality of interfaces associated with the first application or the second application or a plurality of applications associated with the multi-application network.

In one embodiment the first application comprises an application native to the multi-application network while the second application comprises an application that is non-native to the multi-application network and which is accessible via an application programming interface.

In one embodiment the disclosure, may include using a data engine to receive via an interrogator control input field, a first input command from a computing device. The first input command may include a natural language input. For example, the natural language input may include one or more text or character strings associated with spoken and/or written language (e.g., English language, French language, Portuguese language, Chinese language, Japanese language, Korean language, Dutch language, etc.). The methods may also include using the data engine to analyze, based on a data model, the first input command. In one embodiment, the data model comprises a large language model. The large language model, according to some embodiments, comprises an artificial intelligence (AI) model configured to process or otherwise analyze vast amounts of character strings associated with the spoken and/or written language. In some embodiments, the methods include using the data engine to determine, based on the analysis of the first input command, a digital request data object associated with the first input command. The digital request data object may comprise a document or a file that outlines or is associated with one or more of: account data associated with the digital request data object; or parametric data associated with resolving one or more exception events associated with the digital request data object. The methods may also include using the data engine to resolve, based on the analysis of the first input command, the first input command into a data string. The data string may comprise textual data (e.g., string of characters) and/or vocal or auditory data defining or indicating a word, a phrase, or a sentence associated with the written or spoken language. In one embodiment, the data string comprises a sequence of characters including letters, numbers, symbols, and blank spaces as the case may require. The methods may further include using the data engine to determine, based on the digital request data object and the data string, intent data which may be subsequently used to generate a first set of operation recommendations comprising a first operation recommendation and a second operation recommendation. In exemplary embodiments, the methods further include using the data engine to receive, a selection of the first operation recommendation to determine a first application associated with the first operation recommendation. The methods, may further include using the data engine to determine a first application programming interface asso-

5

ciated with the first application and subsequently connecting the first application programming interface to the first application. Furthermore, the methods include using the data engine to execute the first operation recommendation using the first application thereby resulting in a first computing operation result. According to some embodiments, the methods include using the data engine to transmit, to the computing device (e.g., a tablet, a mobile device, etc.), the first computing operation result. The data engine may be further used to receive, using the data engine, a selection of the second operation recommendation. In response to receiving the selection of the second operation recommendation, the data engine may determine a second application, different from the first application, associated with the second operation recommendation. The data engine may be further used to determine the first application programming interface or a second application programming interface associated with the second application and connect, using the first application programming interface or the second application programming interface, to the second application. The data engine may be further used to execute, using the second application, the second operation recommendation, thereby resulting in a second computing operation result different from the first computing operation result. According to one embodiment, the data engine may transmit to the computing device, the second computing operation result.

These and other implementations may each optionally include one or more of the following features. The data engine may be further used to generate, for example, at least context data (e.g., new or updated context data) associated with the digital request data object or the first computing operation result. In one embodiment, the context data indicates one or more of an exception event or a processing stage associated with the digital request data object. Moreover, the data engine may be used to initiate the display of the context data and/or the first computing result associated with the digital request data object on a graphical user interface. The graphical user interface may comprise, for example, a consolidation of a plurality of graphical user interfaces associated with the first application or a plurality of applications associated with the first set of operation recommendations, or a condensation of a plurality of display elements associated with the first application.

The data engine, may be used, according to some embodiments, to automatically format one or more of: the first computing operation result for display on the first graphical user interface based on the context data (e.g., the new or updated context data); or a second set of operation recommendations that are generated based on the first operation recommendation or the context data (e.g., the new or updated context data). Furthermore, the data engine may be used to detect, using the context data, an exception event associated with the digital request data object. Based on detecting the exception event, the data engine may be used to generate a second set of operation recommendations indicating a stage-wise progression of operations that resolve the exception event. In one embodiment, the data engine may transmit the second set of operation recommendations for display on the first computing device. In some embodiments, the digital assistant is associated with the multi-application network such that the multi-application network is configured for multi-application data processing associated with a plurality of domains comprised in a digital processing space. In addition, the first application may comprise one of: an application native to the multi-application network; or an application that is not native to the multi-application network. In some embodiments, the para-

6

metric data referenced above in association with digital request data object comprises one or more identifier data associated with the digital request data object and/or quantitative data associated with the digital request data object and/or exception event data associated with the digital request data object. Furthermore, the data model may be configured to track or assimilate a trajectory of a plurality of input commands including the first input command leading to a selection of specific operation recommendations including the first operation recommendation. Based on the tracking, the data model may be optimized and used by the data engine to recommend a second set of operation recommendations for display on a graphical user interface associated with the first computing device or a second computing device.

## BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings in which like reference numerals are used to refer to similar elements. It is emphasized that various features may not be drawn to scale and the dimensions of various features may be arbitrarily increased or reduced for clarity of discussion. Further, some components may be omitted in certain figures for clarity of discussion.

FIG. 1 is a high level diagram of an exemplary system indicting an implementation of a multi-application network, in accordance with some embodiments of this disclosure.

FIGS. 2 and 3 illustrate exemplary functional and system diagrams, respectively, of a computing environment for implementing a digital assistant in a multi-application network.

FIGS. 4A, 4B, and 5 show exemplary flowcharts for methods, systems/apparatuses, and computer program products that implement a digital assistant in a multi-application network such as the multi-application network of FIG. 1.

FIG. 6 shows an exemplary dynamic single interface associated with the digital assistant of the multi-application network.

## DETAILED DESCRIPTION

### Multi-Application Network

It is increasingly becoming needful to leverage computational tools that automatically recognize relationships among a plurality of disparate data associated with a given multi-application network, and suggest, estimate, or otherwise recommend operations that can be executed to make said disparate data more meaningful, insightful, and readily ingestible or accessible by other computing systems or applications for further processing or analysis. There is therefore a need to develop a digital assistant that can recommend operations based on data relationships in order to eliminate or otherwise minimize time constraints associated with computing operations within the multi-application network. Furthermore, the cost in terms of time, accuracy, and user experience (e.g., navigating multiple similar or dissimilar tools/interfaces) associated with data analysis can affect productivity and/or workflow efficiency, computational or otherwise, within the multi-application network.

Illustrated in FIG. 1 is a high level diagram of an exemplary system 100 indicting an implementation of a multi-application network. In the illustrated implementation, the system 100 may include a cloud server 105 communicatively coupled to a plurality of network systems 130a . . . 130n via a network 110. The system 100 may also include

an endpoint device **125** and cloud storage **113** communicatively coupled via the network **110**. While a single cloud server **105** and a single endpoint device **125** are illustrated, the disclosed principles and techniques could be expanded to include multiple cloud servers, multiple endpoints, and multiple cloud storage devices.

In some embodiments, the cloud server **105** may include a computing device such as a mainframe server, a content server, a communication server, a laptop computer, a desktop computer, a handheld computing device, a smart phone, a wearable computing device, a tablet computing device, a virtual machine, a mobile computing device, a cloud-based computing solution and/or a cloud-based service, and/or the like. The cloud server **105** may include a plurality of computing devices configured to communicate with one another and/or implement the techniques described herein.

The cloud server **105** may include various elements of a computing environment as described in association with the computing environment **200** of FIGS. **2** and **3**. For example, the cloud server **105** may include processing unit **202**, a memory unit **204**, an input/output (I/O) unit **206**, and/or a communication unit **208** which are discussed in association with FIGS. **2** and **3**. The cloud server **105** may further include subunits and/or other modules for performing operations associated with implementing a digital assistant in a multi-application network, registering a digital command in a multi-application network, generating dynamic context data associated with a digital request data object in a multi-application network, curating data associated with a multi-application network, and generating one or more digital records indicating computing operations and state data within a multi-application network. The cloud server may be locally or remotely operated as the case may require.

Turning back to FIG. **1**, the cloud server **105** may include a web server **115**, a data engine **140**, and a web and agent resources **160**. The web server **115**, the data engine **140** and the web and agent resources **160** may be coupled to each other and to the network **110** via one or more signal lines. The one or more signal lines may comprise wired and/or wireless connections.

The web server **115** may include a secure socket layer (SSL) proxy **145** for establishing HTTP-based connectivity **150** between the cloud server **105** and other devices or systems coupled to the network **110**. Other forms of secure connection techniques, such as encryption, may be employed on the web server **115** and across other systems coupled to the network **110**. Additionally, the web server **115** may deliver artifacts (e.g., binary code, instructions, data, etc.) to the data engine **140** either directly via the SSL proxy **145** and/or via the network **110**. Additionally, the web and agent resources **160** of the cloud server **105** may be provided to the endpoint device **125** via the web app **165** on the web server **115**. The web and agent resources **160** may be used to render a web-based graphical interface (GUI) **170** via the browser **155** running on the endpoint device **125**.

The data engine **140** may either be implemented on the cloud server **105** and/or on the endpoint device **125**. The data engine **140** may include one or more instructions or computer logic that are executed by the one or more processors such as processors such as the processors discussed in association with FIGS. **2** and **3**. In particular, the data engine facilitates executing the processing procedures, methods, techniques, and workflows provided in this disclosure. Some embodiments include an iterative refinement of one or more data models (e.g., learning model, large language model) associated with the multi-application network disclosed via feedback loops executed by one or more computing device

processors and/or through other control devices or mechanisms that make determinations regarding optimization of a given action, template, or model.

In some embodiments, the data engine **140** may access an operating system **180** of the endpoint device **125** in order to execute the disclosed techniques on the endpoint device **125**. For instance, the data engine **140** may gain access into the operating system **180** including the system configuration module **185**, the file system **190**, and the system services module **195** in order to execute computing operations associated with implementing a digital assistant in a multi-application network, registering a digital command in a multi-application network, generating dynamic context data associated with a digital request data object in a multi-application network, curating data associated with a multi-application network, and generating one or more digital records indicating computing operations and state data within a multi-application network. The plug-in **175** of the web browser **155** may provide needed downloads that facilitate operations executed by the operating system **180**, the data engine **140**, and/or other applications running on the endpoint device **125**.

The network **110** may include a plurality of networks. For instance, the network **110** may include any wired and/or wireless communication network that facilitates communication between the cloud server **105**, the cloud storage **113**, and the endpoint device **125**. The network **110**, in some instances, may include an Ethernet network, a cellular network, a computer network, the Internet, a wireless fidelity (Wi-Fi) network, a light fidelity (Li-Fi) network, a Bluetooth network, a radio frequency identification (RFID) network, a near-field communication (NFC) network, a laser-based network, a 5G network, and/or the like.

The network systems **130a . . . 130n** may include one or more computing devices or servers, services, or applications that can be accessed by the cloud server **105** and/or the endpoint device **125** and/or the cloud database **113** via the network **110**. In one embodiment, the network systems **130a . . . 130n** comprises third-party applications or services that are native or non-native to either the cloud server **105** and/or the endpoint device **125**. The third-party applications or services, for example, may facilitate executing one or more computing operations associated with resolving an exception event associated with a digital request data object. As further discussed below, the digital request data object may comprise a document or a file outlining one or more of: account data associated with a client request; or parametric data associated with resolving one or more exception events associated with the digital request data object. According to some implementations, the applications or services associated with the network systems **130a . . . 130n** and/or associated with the cloud server **105**, and/or the endpoint device **125** must be registered to activate or otherwise enable their usage in the multi-application network. In such cases, the applications and/or services may be encapsulated in a registration object such that the registration object is enabled or activated for use by the data engine **140** based on one or more of: context data associated with a first user input; device profile data associated with a first interface through which the first user input was received; and user profile data associated with the user providing the first user input. On the flip side, the applications and/or services may be encapsulated in a registration object such that the registration object is deactivated or blocked from usage by data engine **140** based on one or more of: context data associated with a second user input; context data associated with a second input; device profile data associated with a second interface

through which the second input was received; and user profile data associated with a user providing the second input. The first and second user inputs may both be textual or auditory and may comprise a natural language input.

Returning to FIG. 1, the cloud storage **113** may comprise one or more storage devices that store data, information and instructions used by the cloud server **105** and/or the endpoint device **125**. The stored information may include information about users, information about data models (e.g., learning model, an artificial intelligence model, etc.), information about a digital assistant associated with the multi-application network, information associated with a digital request data object, information about analysis operations executed by the data engine **140**, etc. In one embodiment, the one or more storage devices mentioned above in association with the cloud storage **113** can be non-volatile memory or similar permanent storage device and media. For example, the one or more storage devices may include a hard disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, solid state media, or some other mass storage device known in the art for storing information on a more permanent basis. While the cloud storage **113** is shown as being coupled to the cloud server **105** and the endpoint device **125** via the network **110**, the data in the cloud storage **113** may be replicated, in some embodiments, on the cloud server **105** and/or the endpoint device **125**. That is to say that a local copy of the data in the cloud storage **113** may be stored on the cloud server **105** and/or the endpoint device **125**. This local copy may be synched with the cloud storage **113** so that when there are any changes to the information in the cloud storage **113**, the local copy on either the cloud server **105** or the endpoint device **125** is also similarly updated or synched in real-time or in near-real-time to be consistent with the information in the cloud storage **113** and vice versa.

Turning back to FIG. 1, the endpoint device **125** may be a handheld computing device, a smart phone, a tablet, a laptop computer, a desktop computer, a personal digital assistant (PDA), a smart device, a wearable device, a biometric device, a computer server, a virtual server, a virtual machine, a mobile device, and/or a communication server. In some embodiments, the endpoint device **125** may include a plurality of computing devices configured to communicate with one another and/or implement the techniques described in this disclosure. It is appreciated that according to some implementations, the endpoint device may be used by a user to access a digital assistant associated with the multi-application network for executing a plurality of operations associated with digital request data object. The data engine **140** may use the assistant to communicate with the user and to execute a plurality of analysis operations as further discussed below.

The local storage **103**, shown in association with the endpoint device **125**, may include one or more storage devices that store data, information, and instructions used by the endpoint device **125** and/or other devices coupled to the network **110**. The stored information may include various logs/records or event files (e.g., exception event data associated with a digital request data object), security event data, etc. The one or more storage devices discussed above in association with the local database **103** can be non-volatile memory or similar permanent storage device and media. For example, the one or more storage devices may include a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, solid state media, or some

other mass storage device known in the art for storing information on a more permanent basis.

The other elements of the endpoint device **125** are discussed in association with the computing environment **200** of FIGS. 2 and 3. For example, elements such as a processing unit **202**, a memory unit **204**, an input/output (I/O) unit **206**, and/or a communication unit **208** may execute one or more of the modules of endpoint device **125** and/or one or more elements of the cloud server **105** shown in FIG. 1. The endpoint device **125** may also include subunits and/or other computing instances as provided in this disclosure for performing operations associated with digital request data object and/or the multi-application network.

FIGS. 2 and 3 illustrate exemplary functional and system diagrams of a computing environment **200**, according to some embodiments of this disclosure, for implementing a digital assistant in a multi-application network, registering a digital command in a multi-application network, generating dynamic context data associated with a digital request data object in a multi-application network, curating data associated with a multi-application network, and generating one or more digital records indicating computing operations and state data within a multi-application network. Specifically, FIG. 2 provides a functional block diagram of the computing environment **200**, whereas FIG. 3 provides a detailed system diagram of the computing environment **200**.

As seen in FIGS. 2 and 3, the computing environment **200** may include a processing unit **202**, a memory unit **204**, an I/O unit **206**, and a communication unit **208**. The processing unit **202**, the memory unit **204**, the I/O unit **206**, and the communication unit **208** may include one or more subunits for performing operations described in this disclosure. Additionally, each unit and/or subunit may be operatively and/or otherwise communicatively coupled with each other and to the network **110**. The computing environment **200** may be implemented on general-purpose hardware and/or specifically-purposed hardware as the case may be. Importantly, the computing environment **200** and any units and/or subunits of FIGS. 2 and/or 3 may be included in one or more elements of system **100** as described in association with FIG. 1. For example, one or more elements (e.g., units and/or subunits) of the computing environment **200** may be included in the cloud server **105** and/or the endpoint device **125** and/or the network systems **130a** . . . **130n**.

The processing unit **202** may control one or more of the memory unit **204**, the I/O unit **206**, and the communication unit **208** of the computing environment **200**, as well as any included subunits, elements, components, devices, and/or functions performed by the memory unit **204**, I/O unit **206**, and the communication unit **208**. The described sub-elements of the computing environment **200** may also be included in similar fashion in any of the other units and/or devices included in the system **100** of FIG. 1. Additionally, any actions described herein as being performed by a processor may be taken by the processing unit **202** of FIGS. 2 and 3 alone and/or by the processing unit **202** in conjunction with one or more additional processors, units, subunits, elements, components, devices, and/or the like. Further, while one processing unit **202** may be shown in FIGS. 2 and 3, multiple processing units may be present and/or otherwise included in the computing environment **200** or elsewhere in the overall system (e.g., system **100** of FIG. 1). Thus, while instructions may be described as being executed by the processing unit **202** (and/or various subunits of the processing unit **202**), the instructions may be executed simultaneously, serially, and/or otherwise by one or multiple processing units **202** on one or more devices.

## 11

In some embodiments, the processing unit **202** may be implemented as one or more computer processing unit (CPU) chips and/or graphical processing unit (GPU) chips and may include a hardware device capable of executing computer instructions. The processing unit **202** may execute instructions, codes, computer programs, and/or scripts. The instructions, codes, computer programs, and/or scripts may be received from and/or stored in the memory unit **204**, the I/O unit **206**, the communication unit **208**, subunits, and/or elements of the aforementioned units, other devices, and/or computing environments, and/or the like.

In some embodiments, the processing unit **202** may include, among other elements, subunits such as a content management unit **212**, a location determination unit **214**, a graphical processing unit (GPU) **216**, and a resource allocation unit **218**. Each of the aforementioned subunits of the processing unit **202** may be communicatively and/or otherwise operably coupled with each other.

The content management unit **212** may facilitate generation, modification, analysis, transmission, and/or presentation of content. Content may be file content, exception event content, content associated with a digital request data object, content associated with a registration object (e.g., a registration data object associated with registering a command or an application for use by the digital assistant), media content, security event content, or any combination thereof. In some instances, content on which the content management unit **212** may operate includes device information, user interface data, image data, text data, themes, audio data or audio files, video data or video files, documents, and/or the like. Additionally, the content management unit **212** may control the audio-visual environment and/or appearance of application data during execution of various processes (e.g., via web GUI **170** at the endpoint device **125**). In some embodiments, the content management unit **212** may interface with a third-party content server (e.g., third-party content server associated with the network systems **130a** . . . **130n**), and/or specific memory locations for execution of its operations.

The location determination unit **214** may facilitate detection, generation, modification, analysis, transmission, and/or presentation of location information. Location information may include global positioning system (GPS) coordinates, an internet protocol (IP) address, a media access control (MAC) address, geolocation information, a port number, a server number, a proxy name and/or number, device information (e.g., a serial number), an address, a zip code, and/or the like. In some embodiments, the location determination unit **214** may include various sensors, radar, and/or other specifically-purposed hardware elements for the location determination unit **214** to acquire, measure, and/or otherwise transform location information.

The GPU **216** may facilitate generation, modification, analysis, processing, transmission, and/or presentation of content described above, as well as any data described herein. In some embodiments, the GPU **216** may be utilized to render content for presentation on a computing device (e.g., via web GUI **170** at the endpoint device **125**). The GPU **216** may also include multiple GPUs and therefore may be configured to perform and/or execute multiple processes in parallel.

The resource allocation unit **218** may facilitate the determination, monitoring, analysis, and/or allocation of computing resources throughout the computing environment **200** and/or other computing environments. For example, the computing environment may facilitate a high volume of data (e.g., data associated with a digital request data object or a

## 12

registration object), to be processed and analyzed. As such, computing resources of the computing environment **200** used by the processing unit **202**, the memory unit **204**, the I/O unit **206**, and/or the communication unit **208** (and/or any subunit of the aforementioned units) such as processing power, data storage space, network bandwidth, and/or the like may be in high demand at various times during operation. Accordingly, the resource allocation unit **218** may include sensors and/or other specially-purposed hardware for monitoring performance of each unit and/or subunit of the computing environment **200**, as well as hardware for responding to the computing resource needs of each unit and/or subunit. In some embodiments, the resource allocation unit **218** may use computing resources of a second computing environment separate and distinct from the computing environment **200** to facilitate a desired operation. For example, the resource allocation unit **218** may determine a number of simultaneous computing processes and/or requests. The resource allocation unit **218** may also determine that the number of simultaneous computing processes and/or requests meet and/or exceed a predetermined threshold value. Based on this determination, the resource allocation unit **218** may determine an amount of additional computing resources (e.g., processing power, storage space of a particular non-transitory computer-readable memory medium, network bandwidth, and/or the like) required by the processing unit **202**, the memory unit **204**, the I/O unit **206**, the communication unit **208**, and/or any subunit of the aforementioned units for safe and efficient operation of the computing environment while supporting the number of simultaneous computing processes and/or requests. The resource allocation unit **218** may then retrieve, transmit, control, allocate, and/or otherwise distribute determined amount(s) of computing resources to each element (e.g., unit and/or subunit) of the computing environment **200** and/or another computing environment.

The memory unit **204** may be used for storing, recalling, receiving, transmitting, and/or accessing various files and/or data during operation of computing environment **200**. For example, memory unit **204** may be used for storing, recalling, and/or updating exception event information as well as other data associated with, resulting from, and/or generated by any unit, or combination of units and/or subunits of the computing environment **200**. In some embodiments, the memory unit **204** may store instructions, code, and/or data that may be executed by the processing unit **202**. For instance, the memory unit **204** may store code that execute operations associated with one or more units and/or one or more subunits of the computing environment **200**. For example, the memory unit may store code for the processing unit **202**, the I/O unit **206**, the communication unit **208**, and for itself.

Memory unit **204** may include various types of data storage media such as solid state storage media, hard disk storage media, virtual storage media, and/or the like. Memory unit **204** may include dedicated hardware elements such as hard drives and/or servers, as well as software elements such as cloud-based storage drives. In some implementations, memory unit **204** may be a random access memory (RAM) device, a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory, read only memory (ROM) device, and/or various forms of secondary storage. The RAM device may be used to store volatile data and/or to store instructions that may be executed by the processing unit **202**. For example, the instructions stored by the RAM device may be a command, a current operating state of computing envi-

13

ronment **200**, an intended operating state of computing environment **200**, and/or the like. As a further example, data stored in the RAM device of memory unit **204** may include instructions related to various methods and/or functionalities described herein. The ROM device may be a non-volatile memory device that may have a smaller memory capacity than the memory capacity of a secondary storage. The ROM device may be used to store instructions and/or data that may be read during execution of computer instructions. In some embodiments, access to both the RAM device and ROM device may be faster to access than the secondary storage.

Secondary storage may comprise one or more disk drives and/or tape drives and may be used for non-volatile storage of data or as an over-flow data storage device if the RAM device is not large enough to hold all working data. Secondary storage may be used to store programs that may be loaded into the RAM device when such programs are selected for execution. In some embodiments, the memory unit **204** may include one or more databases **310** (shown in FIG. **3**) for storing any data described herein. For example, depending on the implementation, the one or more databases may be used as the local record repository **103** of the endpoint device discussed with reference to FIG. **1**. Additionally or alternatively, one or more secondary databases (e.g., the public record repository **113** discussed with reference to FIG. **1**) located remotely from computing environment **200** may be used and/or accessed by memory unit **204**. In some embodiments, memory unit **204** and/or its subunits may be local to the cloud server **105** and/or the endpoint device **125** and/or remotely located in relation to the cloud server **105** and/or the endpoint device **125**.

Turning back to FIG. **2**, the memory unit **204** may include subunits such as an operating system unit **226**, an application data unit **228**, an application programming interface (API) unit **230**, a content storage unit **232**, data engine **140**, and a cache storage unit **240**. Each of the aforementioned subunits of the memory unit **204** may be communicatively and/or otherwise operably coupled with each other and other units and/or subunits of the computing environment **200**. It is also noted that the memory unit **204** may include other modules, instructions, or code that facilitate the execution of the techniques described. For instance, the memory unit **204** may include one or more modules such as a data engine discussed in association with FIGS. **4A** and **4B**.

The operating system unit **226** may facilitate deployment, storage, access, execution, and/or utilization of an operating system utilized by computing environment **200** and/or any other computing environment described herein. In some embodiments, operating system unit **226** may include various hardware and/or software elements that serve as a structural framework for processing unit **202** to execute various operations described herein. Operating system unit **226** may further store various pieces of information and/or data associated with the operation of the operating system and/or computing environment **200** as a whole, such as a status of computing resources (e.g., processing power, memory availability, resource utilization, and/or the like), runtime information, modules to direct execution of operations described herein, user permissions, security credentials, and/or the like.

The application data unit **228** may facilitate deployment, storage, access, execution, and/or utilization of an application used by computing environment **200** and/or any other computing environment described herein. For example, the endpoint device **125** may be required to download, install, access, and/or otherwise use a software application (e.g., web application **165**) to facilitate implementing a digital

14

assistant in a multi-application network, registering a digital command in a multi-application network, generating dynamic context data associated with a digital request data object in a multi-application network, curating data associated with a multi-application network, and generating one or more digital records indicating computing operations and state data within a multi-application network. As such, the application data unit **228** may store any information and/or data associated with an application. The application data unit **228** may further store various pieces of information and/or data associated with the operation of an application and/or computing environment **200** as a whole, such as status of computing resources (e.g., processing power, memory availability, resource utilization, and/or the like), runtime information, user interfaces, modules to direct execution of operations described herein, user permissions, security credentials, and/or the like.

The API unit **230** may facilitate deployment, storage, access, execution, and/or utilization of information associated with APIs of computing environment **200** and/or any other computing environment described herein. For example, computing environment **200** may include one or more APIs for various devices, applications, units, subunits, elements, and/or other computing environments to communicate with each other and/or utilize the same data. Accordingly, API unit **230** may include API databases containing information that may be accessed and/or utilized by applications, units, subunits, elements, and/or operating systems of other devices and/or computing environments. In some embodiments, each API database may be associated with a customized physical circuit included in memory unit **204** and/or API unit **230**. Additionally, each API database may be public and/or private, and so authentication credentials may be required to access information in an API database. In some embodiments, the API unit **230** may enable the cloud server **105** and the endpoint device **125** to communicate with each other. It is appreciated that the API unit **230** may facilitate accessing, using the data engine **140**, one or more applications or services on the cloud server **105** and/or the network systems **130a** . . . **130n**.

The content storage unit **232** may facilitate deployment, storage, access, and/or utilization of information associated with performance of implementing operations associated with a multi-application network and/or framework processes by computing environment **200** and/or any other computing environment described herein. In some embodiments, content storage unit **232** may communicate with content management unit **212** to receive and/or transmit content files (e.g., media content, digital request data object content, command content, input content, registration object content, etc.).

As previously discussed, the data engine **140** facilitates executing the processing procedures, methods, techniques, and workflows provided in this disclosure. In particular, the data engine **140** may be configured to execute computing operations associated with the disclosed methods, systems/apparatuses, and computer program products.

The cache storage unit **240** may facilitate short-term deployment, storage, access, analysis, and/or utilization of data. In some embodiments, cache storage unit **240** may serve as a short-term storage location for data so that the data stored in cache storage unit **240** may be accessed quickly. In some instances, cache storage unit **240** may include RAM devices and/or other storage media types for quick recall of stored data. Cache storage unit **240** may include a partitioned portion of storage media included in memory unit **204**.



15

The I/O unit **206** may include hardware and/or software elements for the computing environment **200** to receive, transmit, and/or present information useful for performing the disclosed processes. For example, elements of the I/O unit **206** may be used to receive input from a user of the endpoint device **125**. As described herein, I/O unit **206** may include subunits such as an I/O device **242**, an I/O calibration unit **244**, and/or driver **246**.

The I/O device **242** may facilitate the receipt, transmission, processing, presentation, display, input, and/or output of information as a result of executed processes described herein. In some embodiments, the I/O device **242** may include a plurality of I/O devices. In some embodiments, I/O device **242** may include a variety of elements that enable a user to interface with computing environment **200**. For example, I/O device **242** may include a keyboard, a touchscreen, a button, a sensor, a biometric scanner, a laser, a microphone, a camera, and/or another element for receiving and/or collecting input from a user. Additionally and/or alternatively, I/O device **242** may include a display, a screen, a sensor, a vibration mechanism, a light emitting diode (LED), a speaker, a radio frequency identification (RFID) scanner, and/or another element for presenting and/or otherwise outputting data to a user. In some embodiments, the I/O device **242** may communicate with one or more elements of processing unit **202** and/or memory unit **204** to execute operations associated with the disclosed techniques and systems.

The I/O calibration unit **244** may facilitate the calibration of the I/O device **242**. For example, I/O calibration unit **244** may detect and/or determine one or more settings of I/O device **242**, and then adjust and/or modify settings so that the I/O device **242** may operate more efficiently. In some embodiments, I/O calibration unit **244** may use a driver **246** (or multiple drivers) to calibrate I/O device **242**. For example, the driver **246** may include software that is to be installed by I/O calibration unit **244** so that an element of computing environment **200** (or an element of another computing environment) may recognize and/or integrate with I/O device **242** for the processes described herein.

The communication unit **208** may facilitate establishment, maintenance, monitoring, and/or termination of communications between computing environment **200** and other computing environments, third party server systems, and/or the like (e.g., between the cloud server **105** and the endpoint device **125** and/or the network systems **130a** . . . **130n**). Communication unit **208** may also facilitate internal communications between various elements (e.g., units and/or subunits) of computing environment **200**. In some embodiments, communication unit **208** may include a network protocol unit **248**, an API gateway **250**, an encryption engine **252**, and/or a communication device **254**. Communication unit **208** may include hardware and/or other software elements.

The network protocol unit **248** may facilitate establishment, maintenance, and/or termination of a communication connection for computing environment **200** by way of a network. For example, the network protocol unit **248** may detect and/or define a communication protocol required by a particular network and/or network type. Communication protocols used by the network protocol unit **248** may include Wi-Fi protocols, Li-Fi protocols, cellular data network protocols, Bluetooth® protocols, WiMAX protocols, Ethernet protocols, powerline communication (PLC) protocols, and/or the like. In some embodiments, facilitation of communication for computing environment **200** may include transforming and/or translating data from being compatible with

16

a first communication protocol to being compatible with a second communication protocol. In some embodiments, the network protocol unit **248** may determine and/or monitor an amount of data traffic to consequently determine which particular network protocol is to be used for establishing a secure communication connection, transmitting data, and/or performing malware scanning operations and/or other processes described herein.

The API gateway **250** may allow other devices and/or computing environments to access the API unit **230** of the memory unit **204** associated with the computing environment **200**. For example, an endpoint device **125** may access the API unit **230** of the computing environment **200** via the API gateway **250**. In some embodiments, the API gateway **250** may be required to validate user credentials associated with a user of the endpoint device **125** prior to providing access to the API unit **230** to a user. The API gateway **250** may include instructions for the computing environment **200** to communicate with another computing device and/or between elements of the computing environment **200**.

### Exemplary Embodiment

Processing or analyzing data in a multi-application network may involve the use of a plurality of technologies or applications associated one or more domains, sectors, processing stages, and/or workflow stages or sub-stages associated with a digital request data object. According to one embodiment, the digital request data object comprises a file, or a document, or a record, or profile data associated with a user request, or profile data associated with a digital service comprised in or associated with the multi-application network. It is appreciated that the file, the document, record, profile data associated with a user request, or profile data associated with a service comprised in, or associated with the multi-application network outlines or is associated with one or more of: account data associated with the digital request data object; or parametric data associated with resolving one or more exception events associated with the digital request data object. According to one embodiment, the multi-application network may: connect a plurality of users via one or more computer networks such as those discussed above in association with FIGS. 1, 2, and 3; include computational capabilities that improve user experience; include one or more analysis operations comprising workflows/logic associated with one or more applications; and include a machine learning or an artificial intelligence engine or module drives context identification associated with a digital request data object and allows further customized interrogation of the multi-application network based on identified context. In some embodiments, the disclosed multi-application network is scalable, and can condense multiple user interfaces into a single user interface based on the digital context associated with a given digital request data object to allow optimized and otherwise seamless generation of analysis data or computing results associated with the processing stages associated with the digital request data object. Moreover, the analysis data, computing results, or context or intent data associated with the digital request data object may be displayed on a single interface thereby negating the laborious process of a user navigating between a plurality of applications and/or interfaces associated with a plurality of applications generating the analysis data, computing results, or context data. This may be achieved by the user interacting with a digital assistant of the multi-application network such that the digital assistant automatically evaluates user inputs (e.g., textual or audio natural

language inputs) and intelligently engages applications associated with the multi-application network to execute the needed process for the various processing stages associated with resolving one or more exception events associated with the digital request data object. For example, the digital assistant may comprise or be associated with an artificial or machine learning engine which adapts to, or intelligently uses the user inputs by leveraging context data associated with one or more of: a user profile; the user inputs; one or more digital request data objects; or an exception event associated with the one or more digital request data objects.

The exception event, for example, may comprise a disruption in processing a digital request data object at a particular processing stage based on the digital request data object's failure to meet certain requirements or criteria at said stage. For example, some of the requirements may include: a lack of additional input from a user associated with the digital request data object; profile data of the user associated with the digital request data object; a lack of analysis data or computing results data from other applications associated with the multi-application network; a time threshold required to process the digital request data object; profile data associated with a device on which the digital request data object is being processed; credential data associated with a user of the multi-application network; lack of context data associated with the digital request data object; lack of client profile data associated with a client corresponding to the digital request data object.

According to one embodiment, the multi-application network provides users with a functionality to operate on one or more requests (e.g., digital requests data objects) regardless of the status (e.g., exception event status) of said requests in a user-friendly and context-enabled machine learning user interface. The multi-application network can provide a user with relevant data based on a digital context within which the user is interacting or communicating with the digital assistant of the multi-application network. In one embodiment, the multi-application network can generate a context-specific interface based on the type of inquiry or operations associated with inputs being received so that a user does not need to actively navigate to multiple user interfaces or access relevant data processing applications that interpret or otherwise analyze data at a given processing stage of a digital request data object. In some embodiments, an application programming interface (API) such as those discussed in conjunction with FIGS. 2 and 3 facilitates the selection of relevant programs or digital logic for processing data associated with the digital request data object based on the digital context.

According to some embodiments, the multi-application network enables automating exception event detection associated with one or more digital request data objects with little to no user intervention using APIs that access and process stage-specific or domain-specific applications associated with the digital request data object. Results from such processes may be presented together with one or more recommendations on possible operations or workflows to execute to resolve and or track an identified exception on a single graphical user interface associated with the multi-application network. According to one embodiment, an identified exception of the digital request data object indicates a digital event, a digital disruption, a data anomaly, or data condition that needs to be resolved or satisfied for a stage-wise progression of the digital request data object, for example, to move the digital request data object to a different processing stage associated with the digital request data object. In one embodiment, the multi-application net-

work can improve efficiency of operations, computing or otherwise, within the multi-application network by decreasing:

- 1) the cognitive load for teams (e.g., Agile team) using the multi-application network,
- 2) back-office (full-time equivalent FTE) operations associated with the multi-application network by at least 25% and,
- 3) user training time associated with the multi-application network by at least half of the time required to train users without the multi-application network.

Furthermore, the multi-application network can increase cross-domain data fungibility or cross-specialty data fungibility or cross-stage data fungibility associated with the digital request data object.

The disclosed technology beneficially provides a broad set of functionalities by developing, delivering, and releasing distributed and stand-alone solutions for inter-domain, inter-stage data management through the use of a single multi-application network. Furthermore, the disclosed methods and systems provide a conversational architecture to facilitate interactions between a user and the multi-application network using a digital assistant. In particular, the multi-application network can receive natural language queries/inputs associated with a digital request data object from a user, analyze said natural language queries and provide workflow recommendations as well as other responses associated said natural language queries. In some embodiments, the multi-application network can generate intent/context data indicating a digital context associated with the natural language queries and/or associated with a given digital request data object to generate one or more workflow recommendations and/or link a user query/input to one or more digital request data objects and/or applications. It is appreciated that the multi-application network can maintain a digital context associated with a given digital request data object based on one or more of a digital request of a client, a user input from a user, or a digital request data object associated with the user input. According to some embodiment, the multi-application network can curate data based on a profile of a user (e.g., a digital profile of a user), a digital profile of the digital request data object, a digital profile of a computing device being used by the user, location data of a user, and other security protocols associated with the user and/or the digital request data object. It is appreciated that the multi-application network may facilitate implementing a digital assistant, executing registration operations for digital commands, executing context awareness operations, executing operations associated with curating data, and executing operations associated with generating analysis reports or receipts responsive to executing one or more computing operations associated with a digital request data object and/or registration object associated with registering a command.

#### Digital Assistant

The digital assistant of the multi-application network may comprise a machine learning or an artificial intelligence engine module that can receive user input(s) (e.g., textual or auditory natural language inputs) associated with a digital request data object. Following reception of the user input(s), the digital assistant may infer or otherwise analyze said user inputs to generate intent or context data that provides a digital context associated with one or more of the user input taking into consideration the digital request data object associated with the user input, profile data associated with the user entering the user input, exception event data associated with the digital request data object, current or previ-

ously executed operations at one or more processing stages which have been executed on the digital request data object. In some embodiments, the digital assistant interacts with a user via a single user interface such as the one shown in FIG. 6. In particular, the digital assistant can beneficially facilitate seamlessly extracting and/or computationally operating on data associated with a digital request data object within the multi-application network based on context data thereby eliminating the need to process or analyze vast amounts of data that may not be contextually needful or pertinent to a given analysis at a given processing stage of the digital request data object. This advantageously improves user experience by eliminating the need for the user to interact with multiple interfaces and/or computational tools required to provide appropriate analysis data or computation results to the user.

In some embodiments, the digital assistant is implemented based on a microfrontend architecture that uses one or more user-inputs or commands in, for example, natural language to interpret user intent and thereby generate the intent or context data indicating the digital context. According to one embodiment, the context data may control, confirm, or otherwise regulate: the formatting and/or presentation of analysis operation recommendations based on the one or more user inputs; exception event detection and mitigation operations associated with the digital request data object; recommendation of stage-wise progression of operations that resolve exception events associated with the digital request data object; and communication with one or more APIs coupled to one or more native or non-native applications or other third-party applications that execute workflows based on one or more of a user input, a selected analysis operation recommendation, or the context data. In some embodiments, the digital assistant can match one or more keywords in a user input to one or more dynamically configured operations associated with a digital request data object to generate the one or more analysis operation recommendations for selection by the user. Furthermore, based on the context data, the digital assistant can suggest other tasks or data types to the user for selection as the case may require. If the digital assistant makes a suggestion that the user is not interested in, the user can provide other input commands (e.g., natural language input commands) clarifying the context data for the digital assistant to generate pertinent outputs for selection by the user. In such cases, the digital assistant is able to track and/or assimilate a trajectory of user inputs leading to an eventual recommendation selection by the user and at a later date, provide said recommendations for the same or other sets of user inputs associated with the same or different digital request data objects.

#### Exemplary Flowcharts

FIGS. 4A, 4B, and 5 show exemplary flowcharts for methods, systems/apparatuses, and computer program products that implement a digital assistant in a multi-application network such as the multi-application network of FIG. 1. It is appreciated that a data engine stored in a memory device (e.g., memory unit 204 of FIGS. 2 and 3) may cause a computer processor to execute the various processing stages of FIGS. 4A, 4B, and 5. At block 402, the data engine may receive via an interrogator control input field, a first input command from a computing device. According to some embodiments, the first input command comprises a natural language input. For example, the natural language input may include one or more textual or character string data associated with spoken and/or written language (e.g., English language, French language, Portuguese language, Chinese language, Japanese language, Korean language, Dutch lan-

guage). At block 404, the data engine may analyze, using a data model, the first input command. In one embodiment, the data model comprises a large language model. The large language model, according to some embodiments, comprises an artificial intelligence (AI) or a machine learning model configured to process or otherwise analyze vast amounts of character strings associated with the spoken and/or written language. At block 406, the data engine may determine, based on analyzing the first input command, a digital request data object associated with the first input command. As discussed under the exemplary embodiments above, the digital request data object can comprise a file, or a document, or a record, or profile data associated with a user request (e.g., a digital request data object), or profile data associated with a digital service comprised in or associated with the multi-application network. The file, document, record, or profile data associated with the digital request data object, or profile data associated with a service comprised in, or associated with the multi-application network can outline or be associated with one or more of: account data associated with the digital request data object; or parametric data associated with resolving one or more exception events associated with the digital request data object. At block 408, the data engine may resolve, based on analyzing the first input command, the first input command into a data string (e.g., a character string, or one or more words, or one or more phrases). The data string may comprise textual data (e.g., string of characters) and/or vocal or auditory data defining or indicating a word, a phrase, or a sentence associated with written or spoken language (e.g., English language, French language, Portuguese language, Chinese language, Japanese language, Korean language, Dutch language). In one embodiment, the data string comprises a sequence of characters including letters, numbers, symbols, and blank spaces as the case may require. At block 410, the data engine may determine, based on the digital request data object and the data string, intent data. At block 412, the data engine can generate, based on the intent data, a first set of operation (e.g., computing operation) recommendations comprising a first operation recommendation. At block 414, the data engine can receive, a selection of the first operation recommendation. At block 416 of FIG. 4B, the data engine, in response to receiving the selection of the first operation recommendation, can determine a first application associated with the first operation recommendation and which is comprised in a collection of applications associated with the multi-application network. At block 418, the data engine can determine, a first application programming interface associated with the first application. At block 420, the data engine can connect, the first application programming interface, to the first application. At block 422, the data engine can execute, using the first application, the first operation recommendation, thereby resulting in a first computing operation result. At block 424, the data engine can transmit, to the computing device, the first computing operation result.

Turning to FIG. 5, the data engine may, at block 502, receive a selection of the second operation recommendation. At block 504, the data engine may, in response to receiving the selection of the second operation recommendation, determine a second application, different from the first application, associated with the second operation recommendation. At block 506, the data engine may determine the first application programming interface or a second application programming interface associated with the second application. The data engine may, at block 508, connect, using the first application programming interface or the

21

second application programming interface, to the second application. The data engine, at block 510, may execute, using the second application, the second operation recommendation, thereby resulting in a second computing operation result different from the first computing operation result. The data engine may, at block 512, transmit to the computing device, the second computing operation result.

Continuing at block 514 of FIG. 5, the data engine may further generate at least context data (e.g., new or updated context data derived from the intent data) associated with the digital request data object or the first computing operation result based on, for example, the analysis of the first input command or a second input command or the selection of the first computing operation recommendation discussed in association with FIGS. 4A and 4B. In some embodiments, the context data indicates one or more of an exception event or a processing stage associated with the digital request data object. Furthermore, the data engine may initiate, at block 516, display of the context data and/or the first computing result associated with the digital request data object on a graphical user interface. The graphical user interface (e.g., a first graphical user interface) may comprise, for example, a consolidation of a plurality of graphical user interfaces associated with the first application or a plurality of applications associated with the first set of operation recommendations, or a condensation of a plurality of display elements associated with the first application or a plurality of applications associated with the first set of operation recommendations.

In one embodiment, the disclosed techniques may involve receiving a third input (e.g., natural language input), or a fourth input (e.g., natural language input), or a fifth input (e.g., natural language input) and analyze the third input, the fourth input, or the fifth input based on context data or an updated version of the context data associated with a previous first input or a previous second input to generate a third set of operation recommendations or a fourth set of operation recommendations that are displayable, using the digital assistant, on a graphical interface as identifiers (e.g., application identifiers) associated with one or more applications of the multi-application network. It is appreciated that the digital assistant disclosed leverages context data or updated context data associated with a digital request data object to enhance recommending computing operations associated with the digital request data object to a user. This beneficially negates the need to individually locate and/or access applications and/or interact with specific plurality of user interfaces associated with said applications to conduct analysis operations and/or other computing operations associated with the digital request data object.

These and other implementations may each optionally include one or more of the following features. The data engine, may be used, according to some embodiments, to automatically format one or more of: the first computing operation result for display on the first graphical user interface based on the context data (e.g., the new or updated context data); or a second set of operation recommendations having an associated second application that is generated based on the first operation recommendation or the context data (e.g., the new or updated context data). In one embodiment, the first graphical user interface is a single graphical user interface that is configured to display a plurality of analysis data including the first computing operation result. The plurality of analysis data may be generated from analysis operations associated with one or more digital request data objects including the determined digital request data object discussed above in association with block 406 of FIG.

22

4A. In addition, the analysis operations may be executed by one or more native (e.g., comprised in the multi-application network) or non-native (e.g., external to the multi-application network) applications associated with the multi-application network. In some embodiments, the plurality of analysis data or the context data may be displayed on the single graphical user interface instead of a plurality of graphical user interfaces associated with the one or more native or non-native applications associated with the multi-application network.

Furthermore, the data engine may be used to detect, using the context data, an exception event associated with the digital request data object. Based on detecting the exception event, the data engine may be used to generate a second set of operation recommendations indicating a stage-wise progression of operations that resolve the exception event. In one embodiment, the data engine may transmit the second set of operation recommendations for display on the first computing device. In some embodiments, the multi-application network is configured for multi-application or inter-domain or inter-stage data processing associated with: a plurality of domains and/or a plurality of stages associated with the digital request data object; or a plurality of domains comprised in a digital processing space using the first input command or a second input command as the case may require. In addition, the first application may comprise one of: an application native to the multi-application network; or an application that is not native to the multi-application network. In some embodiments, the parametric data referenced above in association with digital request data object comprises one or more identifier data associated with the digital request data object and/or quantitative data associated with the digital request data object and/or exception event data associated with the digital request data object. Furthermore, the data model may be configured to track or assimilate a trajectory of a plurality of input commands including the first input command leading to a selection of specific operation recommendations including the first operation recommendation. In one embodiment, the data model is an artificial intelligence language model or a machine learning language model that executes the tracking. Based on the tracking, the data model may be optimized and used by the data engine, via the digital assistant, to recommend a second set of operation recommendations for display on a graphical user interface associated with the first computing device or a second computing device. In one embodiment, the second set of operation recommendations comprise a second operation having an associated second application that executes the second operation recommendation via a second application programming interface to generate a second computing operation result.

FIG. 6 shows an exemplary single interface (e.g., dynamic single interface) associated with the digital assistant of the multi-application network. The single interface can, according to one embodiment, comprise an interrogator control input field 602 through which one or more textual or auditory inputs may be received. The textual and/or auditory input may comprise a natural language input from the user. In the case of a textual input, a keyboard, mouse, a touchscreen, or some other input device may facilitate the entry of the textual input. The auditory input on the other hand may be received using an input device such as a microphone. It is appreciated that the textual or auditory input can comprise one or more input commands from a user. After a command is received from a user via the interrogator control input field, the digital assistant can analyze, using an artificial intelligence or a machine learning language model, the input

command to determine a digital request data object associated with the input command. In one embodiment, the initial input command comprises identifier data associated with the digital request data object. In other embodiments, the initial input command comprises metadata associated with the digital request data object. In yet other implementations, the initial input command comprises previously generated context data associated with the digital request data object. In any case the identifier data, the metadata, or the context data associated with the digital request data object informs or otherwise guides the digital assistant to retrieve, generate, or display a summary information **604** associated with the digital request data object. It is appreciated that the summary information **604** may include identifier or qualitative data associated with the digital request data object, quantitative data associated with the digital request data object, and/or exception event data associated with the digital request data object.

Furthermore, intent data or context data (e.g., previously generated intent or context data) associated with the digital request data object may also be displayed as shown under the highlights section **606** of the single user interface. According to one embodiment, the data presented under the highlights section **606** of the single user interface comprises a summary of the previously generated context data. Furthermore, the intent or context data may be organic in that it is dynamically updated based on one or more of: the types of computing operations to which the digital request data object is subjected to; a user profile of the user accessing the digital request data object; a device profile of the computing device through which the digital request data object is accessed; location data associated with the computing device through which the digital request data object is accessed; and processing stage data associated with resolving one or more exception events associated with the digital request data.

Turning back to FIG. 6, the digital assistant may generate and/or display, a set of computing operation recommendations **608a**, **608b**, **608c**, **608d**, **608e**, **608f**, etc., based on one or more of the intent or context data and/or parametric data or textual/vocal character data (e.g., specific input textual strings or vocal words) comprised in the input command. In one embodiment, the digital assistant may generate a computation result **610** based on a selection of at least one computing operation recommendation. To achieve this, the digital assistant may use one or more application programming interfaces, to connect, link, or otherwise activate a specific application required to execute the at least one computing operation. According to some embodiments, the digital assistant may generate additional sets of computing operation recommendations **608a** . . . **608f** in conjunction with generating and/or presenting the computing result **610**. It is appreciated that the formatting of the computing result and/or the computing operation recommendations for display on the single graphical user interface may be based on the context data or an updated context data associated with the digital request data object.

It is appreciated that the digital request data object may comprise a file or document indicating a loan request, a request to process medical registration data, a request to process student data associated with an educational institution, a request to process scientific data associated with a research, etc.

This patent application incorporates by reference the following commonly owned applications: (1) U.S. patent application Ser. No. 18/373,771, titled "Methods And Systems For Implementing An Intelligent Digital Assistant In A

Multi-Application Network," filed on Sep. 27, 2023; (2) U.S. patent application Ser. No. 18/373,797, titled "Methods And Systems For Registering A Digital Command In A Multi-Application Network," filed on Sep. 27, 2023; (3) U.S. patent application Ser. No. 18/373,813, titled "Methods And Systems For Generating Dynamic Context Data Associated With A Digital Request Data Object In A Multi-Application Network," filed on Sep. 27, 2023; (4) U.S. patent application Ser. No. 18/373,822, titled "Methods And Systems For Curating Data In A Multi-Application Network," filed on Sep. 27, 2023; and (5) U.S. patent application Ser. No. 18/373,830, titled "Methods And Systems For Generating Digital Records Indicating Computing Operations And State Data In A Multi-Application Network," filed on Sep. 27, 2023.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to explain the principles of the disclosed subject-matter and its practical applications, to thereby enable others skilled in the art to use the technology disclosed and various embodiments with various modifications as are suited to the particular use contemplated.

It is appreciated that the term optimize/optimal and its variants (e.g., efficient or optimally) may simply indicate improving, rather than the ultimate form of 'perfection' or the like.

Furthermore, the functions or operations described above can be implemented in digital electronic circuitry, in computer software, firmware or hardware. In particular, the disclosed techniques can be implemented using one or more computer program products. The computer program products, in some embodiments, comprises non-transitory computer-readable media comprising code configured to execute the disclosed approach. Programmable processors and computers can be included in or packaged as mobile devices according to some embodiments. The processes and logic flows can be performed by one or more programmable processors and by one or more programmable logic circuitry. General and special purpose computing devices and storage devices can be interconnected through communication networks.

It will also be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another. For example, a first object or step could be termed a second object or step, and, similarly, a second object or step could be termed a first object or step, without departing from the scope of the invention. The first object or step, and the second object or step, are both objects or steps, respectively, but they are not to be considered the same object or step.

The terminology used in the description herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used in the description of the invention and the appended claims, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any possible combination of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises" and/or "comprising," when used in this specification,

25

specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context.

Those with skill in the art will appreciate that while some terms in this disclosure may refer to absolutes, e.g., all source receiver traces, each of a plurality of objects, etc., the methods and techniques disclosed herein may also be performed on fewer than all of a given thing, e.g., performed on one or more components and/or performed on one or more source receiver traces. Accordingly, in instances in the disclosure where an absolute is used, the disclosure may also be interpreted to be referring to a subset.

What is claimed is:

1. A method for implementing a digital assistant computing operation in a multi-application network, the method comprising:

receiving, using one or more computing device processors, using an interrogator control input field, a first input command from a computing device;

analyzing, using the one or more computing device processors, the first input command;

resolving, using the one or more computing device processors and based on the first input command, the first input command into at least partially resolved data;

determining, using the one or more computing device processors, based on the first input command, a digital request data object associated with the first input command;

determining, using the one or more computing device processors, based on the digital request data object, intent data;

generating, using the one or more computing device processors and based on the intent data, a first set of operation recommendations comprising a first operation recommendation and a second operation recommendation;

receiving, using the one or more computing device processors, a first selection of the first operation recommendation;

in response to the receiving the first selection of the first operation recommendation, determining, using the one or more computing device processors, a first application function associated with the first operation recommendation;

determining or accessing, using the one or more computing device processors, a first application programming interface associated with the first application function;

connecting, using the one or more computing device processors and the first application programming interface, to the first application function;

executing, using the one or more computing device processors and the first application function, the first operation recommendation, thereby resulting in a first computing operation result;

transmitting, using the one or more computing device processors, to the computing device, the first computing operation result;

receiving, using the one or more computing device processors, a second selection of the second operation recommendation;

in response to receiving the second selection of the second operation recommendation, determining, using the one

26

or more computing device processors, a second application function, different from the first application function, associated with the second operation recommendation;

determining or accessing, using the one or more computing device processors, the first application programming interface or a second application programming interface associated with the second application function;

connecting, using the one or more computing device processors and the first application programming interface or the second application programming interface, to the second application function;

executing, using the one or more computing device processors and the second application function, the second operation recommendation, thereby resulting in a second computing operation result different from the first computing operation result;

transmitting, using the one or more computing device processors, to the computing device, the second computing operation result;

determining, using context data, an exception event associated with the digital request data object;

generating, based on the exception event, a second set of operation recommendations for resolving the exception event;

transmitting, the second set of operation recommendations;

executing, using the one or more computing device processors, a third operation recommendation comprised in the second set of operation recommendations, wherein the third operation recommendation is based on resolving the exception event associated with the digital request data object;

receiving, using the one or more computing device processors, a first user input, wherein the first user input comprises first feedback associated with the second set of operation recommendations;

modifying, using the one or more computing device processors, based on the first user input, the second set of operation recommendations, thereby generating a modified second set of operation recommendations; and

storing, using the one or more computing device processors, at a first time, the second set of operation recommendations, the first user input, and the modified second set of operation recommendations.

2. The method of claim 1, further comprising:

generating the context data, the context data being associated with the digital request data object, the context data indicating one or more exception events or a processing stage associated with the digital request data object; and

initiating, using the one or more computing device processors, display of the context data associated with the digital request data object on a first graphical user interface.

3. The method of claim 1, further comprising automatically formatting or arranging, using the one or more computing device processors, one or more of:

the first computing operation result for display on a first graphical user interface,

the first set of operation recommendations for display on the first graphical user interface or a second graphical user interface,

27

the second computing operation result for display on the first graphical user interface, the second graphical user interface, or a third graphical user interface, or the second set of operation recommendations for display on the first graphical user interface, the second graphical user interface, the third graphical user interface, or a fourth graphical user interface.

4. The method of claim 1, wherein at least one of:

the first application function is associated with or comprised in a first application,

the second application function is associated with or comprised in the first application or a second application, or

the first input command comprises a natural language input.

5. The method of claim 1, further comprising improving, based on a learning computing operation, at least one of the first operation recommendation and the second operation recommendation.

6. The method of claim 1, wherein the multi-application network is configured for inter-domain or multi-application function data processing associated with the digital request data object using the first input command or a second input command.

7. The method of claim 1, wherein the first application function comprises at least one of:

a third application function that is native to the multi-application network, or

a fourth application function that is not native to the multi-application network.

8. The method of claim 1, wherein the digital request data object comprises a document or a file comprising at least one of:

account data associated with the digital request data object, or

parametric data associated with the digital request data object.

9. The method of claim 8, wherein the parametric data comprises at least one of:

identifier data associated with the digital request data object,

quantitative data associated with the digital request data object,

first exception event data associated with the digital request data object, or

resolution data associated with second exception event data associated with the digital request data object.

10. The method of claim 1, wherein a data model associated with the digital assistant computing operation is configured to:

track or assimilate a trajectory of a plurality of input commands including the first input command leading to a selection of specific operation recommendations including the first operation recommendation; and

recommend the second set of operation recommendations for display on a graphical user interface associated with the computing device, wherein the second set of operation recommendations comprises a second operation having an associated second application function that executes the second operation recommendation via the second application programming interface or a third application programming interface to generate the second computing operation result.

11. The method of claim 1, wherein a data model associated with the digital assistant computing operation comprises a large language model.

28

12. An apparatus for implementing a digital assistant computing operation in a multi-application network, the apparatus comprising:

one or more computing system processors; and

memory storing instructions that, when executed by the one or more computing system processors, causes the apparatus to:

receive, using an interrogator control input field, an input command from a computing device;

analyze, the input command,

resolve, based on the input command, the input command into partially resolved data,

determine, based on the input command, a digital request data object associated with the input command,

determine, based on the digital request data object, intent data,

generate, based on the intent data, a first set of operation recommendations comprising a first operation recommendation and a second operation recommendation,

receive a first selection of the first operation recommendation,

in response to the receive the first selection of the first operation recommendation, determine a first application function associated with the first operation recommendation,

determine or access a first application programming interface associated with the first application function,

connect, using the first application programming interface, to the first application function,

execute, using the first application function, the first operation recommendation, thereby resulting in a first computing operation result,

transmit, to the computing device, the first computing operation result,

receive a second selection of the second operation recommendation,

in response to receiving the second selection of the second operation recommendation, determine a second application function, different from the first application function, associated with the second operation recommendation,

determine or access the first application programming interface or a second application programming interface associated with the second application function,

connect, using the first application programming interface, to the second application function,

execute the second operation recommendation, thereby resulting in a second computing operation result different from the first computing operation result,

transmit, to the computing device, the second computing operation result,

determine, using context data, an exception event associated with the digital request data object,

generate, based on the exception event, a second set of operation recommendations for resolving the exception event,

transmit the second set of operation recommendations, execute a third operation recommendation comprised in the second set of operation recommendations, wherein the third operation recommendation is based on resolving the exception event associated with the digital request data object,

receive a first user input, wherein the first user input comprises first feedback associated with the second set of operation recommendations;

modify, based on the first user input, the second set of operation recommendations, thereby generating a modified second set of operation recommendations; and

store, at a first time, the second set of operation recommendations, the first user input, and the modified second set of operation recommendations.

13. The apparatus of claim 12, wherein the one or more computing system processors execute the instructions to:

generate at least the context data associated with the digital request data object, the context data indicating one or more exception events or a processing stage associated with the digital request data object; and

initiate display of the context data associated with the digital request data object on a graphical user interface comprising a consolidation of a plurality of graphical user interfaces associated with the first application function or a plurality of application functions associated with the first set of operation recommendations.

14. The apparatus of claim 13, wherein:

the graphical user interface is a single graphical user interface that is configured to display a plurality of analysis data including the first computing operation result;

the plurality of analysis data is generated from analysis operations associated with one or more digital request data objects including the digital request data object;

the analysis operations are executed by one or more native or non-native application functions associated with the multi-application network; and

the plurality of analysis data or the context data are displayed on the single graphical user interface instead of the plurality of graphical user interfaces associated with the first application function or the plurality of application functions associated with the first set of operation recommendations.

15. The apparatus of claim 13, wherein the one or more computing system processors execute the instructions to automatically format one or more of:

the first computing operation result for display on the graphical user interface based on the context data, or

the second set of operation recommendations that is generated based on the first operation recommendation or the context data.

16. The apparatus of claim 12, wherein the first application function is associated with or comprised in a first application, and wherein the second application function is associated with or comprised in the first application or a second application.

17. The apparatus of claim 12, wherein the digital request data object comprises a document or a file outlining one or more of:

account data associated with the digital request data object, or

parametric data associated with resolving one or more exception events associated with the digital request data object.

18. The apparatus of claim 17, wherein the parametric data comprises one or more of:

identifier data associated with the digital request data object,

quantitative data associated with the digital request data object, or

exception event data associated with the digital request data object.

19. The apparatus of claim 12, wherein a data model associated with the digital assistant computing operation is one of an artificial intelligence language model or a machine learning language model.

20. The method of claim 1, wherein at least one of: the first input command is analyzed using a data model, the second set of operation recommendations comprises a first progression of operations for resolving the exception event, the second set of operation recommendations comprises a stage-wise progression of operations for resolving the exception event.

21. The method of claim 1, further comprising:

determining, using second context data, at a second time, a second exception event associated with a second digital request data object; and

transmitting, using the one or more computing device processors, based on at least a partial similarity of the second exception event associated with the second digital request data object and the exception event associated with the digital request data object, the modified second set of operation recommendations.

22. The method of claim 1, further comprising:

initiating display of, using the one or more computing device processors, at least one of: the first set of operation recommendations, the first computing operation result, the second computing operation result, the second set of operation recommendations, or a third computing operation result associated with the executing the third operation recommendation;

receiving, using the one or more computing device processors, a second user input, wherein the second user input comprises second feedback associated with the at least one of: the first set of operation recommendations, the first computing operation result, the second computing operation result, the second set of operation recommendations, or the third computing operation result; and

modifying, using the one or more computing device processors, based on the second user input, the at least one of: the first set of operation recommendations, the first computing operation result, the second computing operation result, the second set of operation recommendations, or the third computing operation result.

23. A method for implementing a digital assistant computing operation in a multi-application network, the method comprising:

receiving, using one or more computing device processors, using an interrogator control input field, a natural language input command from a computing device; analyzing, using the one or more computing device processors, using a language model, the natural language input command;

resolving, using the one or more computing device processors and based on the natural language input command, the natural language input command into at least partially resolved data;

determining, using the one or more computing device processors, based on the natural language input command, a digital request data object associated with the natural language input command;

determining, using the one or more computing device processors, based on the digital request data object, intent data;

generating, using the one or more computing device processors and based on the intent data, a set of



## 31

operation recommendations comprising a first operation recommendation and a second operation recommendation;

receiving, using the one or more computing device processors, a first selection of the first operation recommendation;

in response to receiving the first selection of the first operation recommendation, determining, using the one or more computing device processors, a first application function associated with the first operation recommendation and which is comprised in a collection of application functions associated with the multi-application network;

determining or accessing, using the one or more computing device processors, a first application programming interface associated with the first application function;

connecting, using the one or more computing device processors and the first application programming interface, to the first application function;

executing, using the one or more computing device processors and the first application function, the first operation recommendation, thereby resulting in a first computing operation result;

transmitting, using the one or more computing device processors, to the computing device, the first computing operation result;

receiving, using the one or more computing device processors, a second selection of the second operation recommendation;

in response to receiving the second selection of the second operation recommendation, determining, using the one or more computing device processors, a second application function, different from the first application function, associated with the second operation recommendation;

determining or accessing, using the one or more computing device processors, the first application programming interface or a second application programming interface associated with the second application function;

## 32

connecting, using the one or more computing device processors and the first application programming interface or the second application programming interface, to the second application function;

executing, using the one or more computing device processors and the second application function, the second operation recommendation, thereby resulting in a second computing operation result different from the first computing operation result;

transmitting, using the one or more computing device processors, to the computing device, the second computing operation result;

determining, using context data, an exception event associated with the digital request data object;

generating, based on the determining the exception event, a second set of operation recommendations for resolving the exception event;

transmitting, the second set of operation recommendations; executing, using the one or more computing device processors, a third operation recommendation comprised in the second set of operation recommendations, wherein the third operation recommendation comprises or is based on resolving the exception event associated with the digital request data object;

receiving, using the one or more computing device processors, a first user input, wherein the first user input comprises first feedback associated with the second set of operation recommendations;

modifying, using the one or more computing device processors, based on the first user input, the second set of operation recommendations, thereby generating a modified second set of operation recommendations; and

storing, using the one or more computing device processors, at a first time, the second set of operation recommendations, the first user input, and the modified second set of operation recommendations.

\* \* \* \* \*